

AIM-ALPHA documentation

Agricultural Land-use Partial Equilibrium Model

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Abstract

This document describes the model structure and system of equations for AIM-ALPHA v1.1.0. AIM-ALPHA is a one-year step sequential dynamic partial equilibrium model targeting agricultural markets and land use. It performs future projections considering 166 countries on the demand side, 400 production units on the supply side, 23 agricultural commodities, and major land-use categories such as cropland, pasture, forest, and other natural land. This model is anticipated to be applicable to research including analyses of food security issues, climate change impacts, evaluations of climate change mitigation measures, and linkages to external assessments of biodiversity impacts related to land use.

Keywords: Integrated assessment model, partial equilibrium, agricultural markets, land use, climate change mitigation

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1 Introduction

This document describes the model structure and system of equations of AIM-ALPHA (the Asia-Pacific Integrated Model - Agricultural Land-use Partial equilibrium model for Harmonized Analysis), a partial equilibrium model for agriculture and land use. The document is organized as follows:

- 1) Model Structure ([Section 2](#))
- 2) System of Equations ([Sections 3 and 4](#))
- 3) Data Processing ([Section 5](#))
- 4) Parameter Estimation ([Section 6](#))
- 5) Model Calculation Flow and Execution Method ([Section 7](#))

This document aims to enhance transparency and reproducibility of the model's analytical components, provide clear instructions for model execution and configuration, facilitate linkage with other models, and support further applications in global and national assessments of sustainable land-use transitions.

2 Model Structure

This section provides an overview of the AIM-ALPHA model. AIM-ALPHA is a recursive-dynamic partial equilibrium model focusing on agricultural markets and land use, operating with an annual time step for the time horizon from 2015 to 2100. The model is formulated as a mixed complementarity problem (MCP) and handles approximately 200,000 variables. The basic model structure is shown in Fig. 1.

In terms of spatial resolution, the demand side covers 166 countries globally, while the production side consists of 400 production units. These production units are defined by overlaying national boundaries [1] with FAO's 230 major river basin boundaries [2]. Figure 2 illustrates the geographical coverage of the countries and production units.

Food production is categorized into 13 primary crops, 6 livestock products, and 4 processed goods. Table 1 lists the food commodity classifications used in the model. The production functions are represented by Leontief-type functions, where each good is produced by combining cropland, feed and pasture, and raw materials as inputs. Other factors such as labor, capital, and energy are aggregated into a single production factor.

Primary crops and livestock products that require land are differentiated by production unit, whereas processed goods, which do not require land inputs, are modeled at the level of the 166 countries. Land use within each production unit is allocated according to a multi-nested logit function that balances total land area, and land shares are determined by relative land prices. The non-pasture land nest includes cropland, forest, other natural land, and afforested land. Protected areas are represented as minimum areas for the corresponding land classes, while the agro-ecological suitability constraint limits cropland expansion.

Food demand is driven by population and GDP, with income elasticity considered for future changes. Food prices are determined endogenously so that supply and demand are balanced in the domestic market. Within each country, the balance between production, domestic demand, and net trade (imports minus exports) is maintained. In the global market, total world imports and exports are balanced after accounting for a calibrated trade residual.

AIM-ALPHA can be linked with other AIM models. AIM-Hub provides socioeconomic, climate-policy, energy-crop, and carbon-price drivers; AFOLUB can provide land-based mitigation assumptions; and AIM-PLUM can return changes in basin-level crop yields resulting from spatial land-use allocation [3, 4, 5].

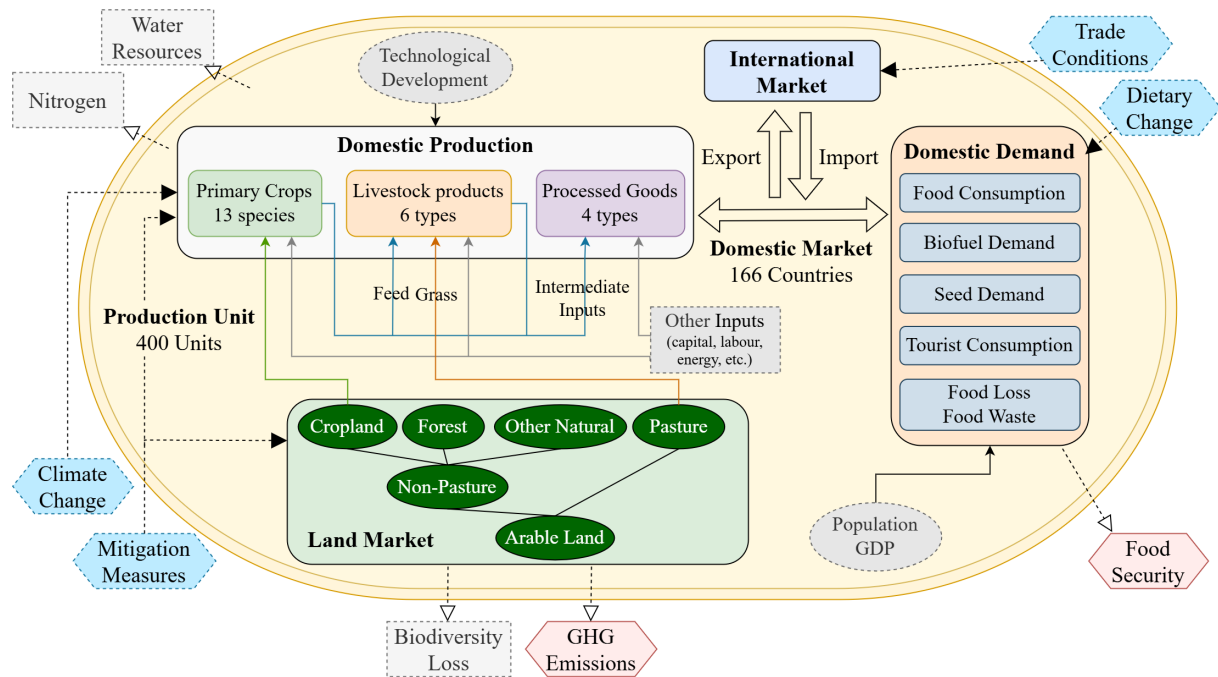


Figure 1: Overview of AIM-ALPHA

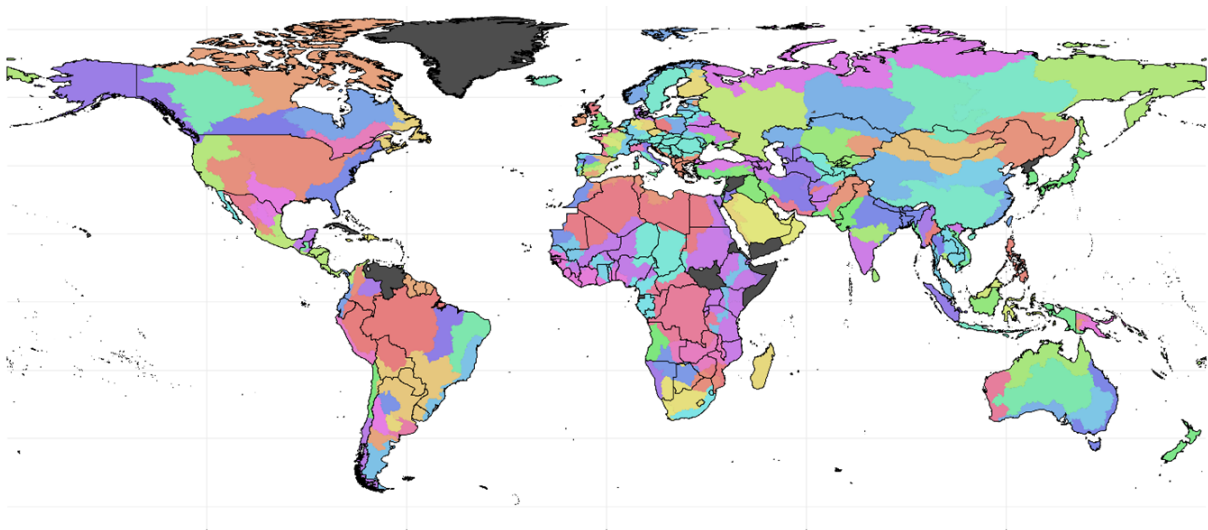


Figure 2: Geographical coverage of countries (166) and production units (400).

Table 1: Classification of food commodities

Category	Commodity	Model code
Primary crops	Wheat	wht
	Rice	rce
	Maize	mze
	Other grains	crl
	Starchy roots	str
	Sugar crops	sgr
	Legumes	pls
	Nuts	nut
	Oilseeds	ocr
	Vegetables	vgt
	Fruits	frt
	Stimulants	stm
	Spices	spc
Livestock products	Beef	cmt
	Sheep and goat meat	rmt
	Poultry meat	pmt
	Other meat	omt
	Eggs	egg
	Raw milk	mlk
Processed products	Sugar and sweeteners	swt
	Vegetable oils	vol
	Alcoholic beverages	alc
	Dairy products	dai

3 Variables and Parameters

This section describes the variables and parameters used in the model.

3.1 Indices

Table 2 lists the indices used in the model.

Table 2: List of indices

Index	Description	Dimension	Notes
cty	Country	166	
bas	Production unit	400	Unique country–river-basin unit
reg	Macro region	17	Same as the regional classification in AIM-Hub [3]. See Section
c(a)	Food commodities (production activities)	23	See Table 1
a_cro	Primary crops	13	
a_liv	Livestock products	6	
a_pro	Processed products	4	
c_fed	Feed inputs	10	
c_int	Intermediate inputs	8	
c_bio	Biofuel crops	3	
wat	Crop production technology	2	Irrigated / rainfed
cz	Climate zone	4	Tropical, temperate, arid, hyper-arid
sys	Livestock production system	2	Pasture-based, mixed farming
lu	Land-use class	7	cro, pas, primf, secdf, gra, pastr, range
lu_pa	Protected land-use class	5	primf, secdf, gra, pastr, range
lu_pa_pas	Protected pasture class	2	pastr, range

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Table 2: List of indices (continued)

Index	Description	Dimension	Notes
src	GHG emission sources	14	Enteric fermentation, manure management, rice cultivation, synthetic fertilizer application, organic fertilizer application, manure on pasture, cultivation of organic soils, crop residues, crop residue burning, grassland burning, forests, natural wildfires, deforestation, afforestation
gas	Gas species	3	CO ₂ , CH ₄ , N ₂ O
age	Forest age	100	1 to 100 years

3.2 Endogenous Variables

Table 3 lists the endogenous variables used in the model.

Table 3: List of endogenous variables

Variable	Description	Unit
QD	Domestic demand	1000t
QDFood	Food consumption demand	1000t
QDFed	Feed demand	1000t
QDInt	Intermediate input demand for processed goods	1000t
QDBio	Biofuel demand	1000t
QDSeed	Seed demand	1000t
QDTour	Tourist consumption demand	1000t
QDLoss	Food loss amount	1000t
QDWaste	Food waste amount	1000t
QDOth	Other demand	1000t
QFeed	Feed input	1000t
QInt	Raw material input	1000t
QS	Domestic production	1000t
QSBas	Production by river basin	1000t
QSWat	Crop production by technology	1000t
QSSys	Livestock production by system	1000t
QM	Imports	1000t
QE	Exports	1000t
NT	Net trade	1000t

Continued on next page

Table 3: List of endogenous variables (continued)

Variable	Description	Unit
PP	Producer price	\$/t
PC	Consumer price	\$/t
PB	Border price	\$/t
PM	Import price	\$/t
PE	Export price	\$/t
PW	World price	\$/t
QGrass	Grass utilization	1000t
YLiv	Livestock yield per hectare	t/ha
NAnimal	Number of livestock	1000 head
COBas	Unit cost of other production inputs by basin	\$/t
CO Cty	Unit cost of other production inputs by country	\$/t
QLVeg	Vegetation area	1000ha
QLPas	Total pasture area	1000ha
QLPasGrz	Grazed pasture area	1000ha
QLPasUng	Ungrazed pasture area	1000ha
QLNpas	Non-pasture vegetation area	1000ha
QLCro	Cropland area	1000ha
QLFrs	Forest area	1000ha
QLPrimf	Primary forest area	1000ha
QLSecdf	Secondary forest area	1000ha
QLNat	Other natural land area	1000ha
QLAfr	Afforested land area	1000ha
QLCroCul	Harvested cropland area	1000ha
QLCroFal	Fallow cropland area	1000ha
QL	Land allocated to a production activity	1000ha
PLPas	Pasture land price (area-weighted average)	\$/ha
PLPasGrz	Grazed pasture land price	\$/ha
PLPasUng	Ungrazed pasture land price	\$/ha
PLNpas	Non-pasture land price	\$/ha
PLCro	Cropland price	\$/ha
PLFrs	Forest price	\$/ha
PLAfr	Afforested land price	\$/ha
PLNat	Other natural land price	\$/ha
PLPrimf	Primary forest land price	\$/ha
PLSecdf	Secondary forest land price	\$/ha
PLCroCul	Harvested cropland price	\$/ha
PL	Land price	\$/ha
PLEne	Energy crop land price	\$/ha

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Table 3: List of endogenous variables (continued)

Variable	Description	Unit
GHG	GHG emissions from agriculture sector	ktCO ₂ -eq/yr
GHGLU	GHG emissions from land-use sector	ktCO ₂ -eq/yr
RRed	Emission reduction rate	-
CAbt	Abatement cost	\$/t
CEmis	Unit residual-emission charge	\$/t
MuBasMin	Complementary variable for basin production	-
MuImpMin	Complementary variable for imports	-
MuPrimfCap	Complementary variable for primary forest expansion	-
MuSecdfCap	Complementary variable for secondary forest expansion	-
MuAfrCap	Complementary variable for the annual afforestation limit	-
MuPrimfProt	Complementary variable for protected primary forest	-
MuSecdfProt	Complementary variable for protected secondary forest	-
MuNatProt	Complementary variable for protected other natural land	-
MuPasProt	Complementary variable for protected pasture	-
MuCroCap	Complementary variable for the cropland constraint	-

3.3 Exogenous Variables

Table 4 lists the exogenous variables used in the model. For data sources, see Section 5.

Table 4: List of exogenous variables

Variable	Description	Data Source
qdFood0	Food consumption demand in the base year	FAOSTAT
qdBio0	Biofuel demand in the base year	FAOSTAT
qdSeed0	Seed demand in the base year	FAOSTAT
qdTour0	Tourist consumption demand in the base year	FAOSTAT
qdLoss0	Food loss in the base year	FAOSTAT
qdOth0	Other demand in the base year	FAOSTAT
qsPrev	One-period-lagged domestic production at the current substep	FAOSTAT
qsBasPrev	One-period-lagged basin production at the current substep	GAEZ v4
qdEne	Demand for dedicated energy crops	AIM-Hub

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Table 4: List of exogenous variables (continued)

Variable	Description	Data Source
qStock	Stock change plus the base-year statistical calibration adjustment	FAOSTAT FBS
qRes	Derived residual of the global international trade balance	FAOSTAT FBS
lossRate	Food loss rate	FAOSTAT
wasteRate	Food waste rate	FAO (2011)
gdpPc	GDP per capita	SSP 3.0 [6]
gdpPc0	GDP per capita in the base year	World Bank
pop	Population	SSP 3.0 [7]
pop0	Population in the base year	World Bank
mmj	Market margin	GTAP 10
pse	Producer Support Estimate (PSE)	GTAP 10
cse	Consumer Support Estimate (CSE)	GTAP 10
cse0	CSE in the base year	GTAP 10
tm	Import tariff rate	GTAP 10
mmm	Import margin rate	GTAP 10
te	Export tariff rate	GTAP 10
mme	Export margin rate	GTAP 10
yld0	Crop yield in the base year	FAOSTAT, GAEZ v4; H08 for energy crops
yld	Annual crop yield	Model calculation
yldTech	Yield change factor due to technological progress	IMPACT
yldChg	Yield change factor due to climate change impacts	GAEZ v4
yldPlum	Optional basin-level yield adjustment from spatial crop allocation	AIM-PLUM
croInt	Cropping intensity	MAPSPAM
yGrass	Grass yield	GAEZ v4
animalProdRate	Slaughtering / milking rate	FAOSTAT
feedReq	Feed requirement per unit of livestock output	Herrero et al. (2018)
feedConcShr	Share of concentrate feed	Herrero et al. (2018)
feedGrainShr	Share of grain feed	FAOSTAT
sysShr	Annual livestock system share	Model calculation
feedCoef	Annual feed input coefficient	Model calculation
animalWt	Livestock body weight	FAOSTAT

Continued on next page

Table 4: List of exogenous variables (continued)

Variable	Description	Data Source
nAnimalRes	Number of non-productive live-stock	FAOSTAT
coBas0	Unit cost of other production inputs by basin in the base year	FAOSTAT, GTAP 10
coCty0	Unit cost of other production inputs by country in the base year	FAOSTAT, GTAP 10
qlTot	Total land area	LUH2
ql0th	Ice/water area	LUH2
qlUrb	Urban area	LUH2; optional SEDAC extension
qlProt	Protected land area by land-use class	LUH2, WDPA
qlProtPas	Protected pasture and rangeland area	LUH2, WDPA
plFrs0	Forest land price in the base year	LUH2, GTAP 10
plPrimf0	Primary forest land price in the base year	LUH2, GTAP 10
plSecdf0	Secondary forest land price in the base year	LUH2, GTAP 10
plNatNet	Price of other natural land before the protection constraint	LUH2, GTAP 10
plAfrRaw	Afforested land price before the annual planting constraint	AIM-PLUM, VISIT
plCroFal0	Fallow land price in the base year	LUH2, GTAP 10
carbonStock	Forest carbon stock by production unit	AIM-PLUM, VISIT
carbonStockNat	Carbon stock of other natural land by production unit	AIM-PLUM, VISIT
carbonSeq	Carbon sequestration by production unit and forest age	AIM-PLUM, VISIT
ef	GHG emission factor in agriculture sector	FAOSTAT
efLUCro	Land-use GHG emission factor for cropland	FAOSTAT
efLUFrs	Land-use GHG emission factor for forest area	FAOSTAT
efLUPas	Land-use GHG emission factor for pasture area	FAOSTAT

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Table 4: List of exogenous variables (continued)

Variable	Description	Data Source
efLUNat	Land-use GHG emission factor for other natural land	FAOSTAT
pCar	Carbon price	AIM-Hub
mrp	Maximum reduction potential of abatement technologies	Harmsen et al. (2019)

3.4 Parameters

Table 5 lists the parameters used in the model.

Table 5: List of parameters

Parameter	Description	Notes
elaInc	Income elasticity of consumption in the current model year	Derived from the Food Demand Model estimates. See Section .
elaP	Price elasticity of consumption	Estimated by regression based on Muhammad et al. (2011)
intCoef	Intermediate input coefficient	
elaCO	Elasticity of other production factors	
δ	Coefficient parameter of the logit function	
ρ	Exponent parameter of the logit function	For land use, adopted from GCAM [8]; for production systems, estimated by regression based on Herrero et al. (2018)
resElaP	Price-elasticity regression residual	
watShr	Share of primary-crop production technologies	
czShr	Climate zone share	
grassUseRate	Grass (pasture) utilization rate (base-year pasture split)	
intCostCal	Intermediate-input-cost calibration coefficient	
feedCostCal	Feed-cost calibration coefficient	
pbRes	Residual between consumer price and border price	
pmRes	Residual between import price and world price	

Continued on next page

Table 5: List of parameters (continued)

Parameter	Description	Notes
peRes	Residual between export price and world price	
exr	Exchange rate	
dcffrs	Discount factor for forest carbon value	Introduces the discounted carbon value of forests, reflecting a 60-year forest lifetime and a 5% interest rate.
afrCost	Afforestation cost	Cost of establishing new forest land, assumed at 250 (100 US\$/acre; 1 acre = 0.4 ha) based on Gorte R.W. (2009).
qlAfrAge	Afforested area by age class	Records annual afforestation by stand age.
σ	Elasticity of adoption of abatement technologies	Estimated by regression based on Harm-sen et al. (2019)

4 System of Equations

This section presents the mathematical formulation of the model and its explanation. In the following, **endogenous variables** begin with uppercase letters, while **parameters** begin with lowercase letters or use Greek notation. An overline identifies an exogenous or predetermined value.

4.1 Demand Block

This block describes the equations related to demand determination. In the demand block, food demand is determined by factors such as prices, population, and GDP.

Total Demand

$$QD_{c,cty} = QDFood_{c,cty} + QDFed_{c,cty} + QDInt_{c,cty} + QDBio_{c,cty} \\ + QDSeed_{c,cty} + QDTour_{c,cty} + QDLoss_{c,cty} + QDWaste_{c,cty} + QDOth_{c,cty} \quad (QDEq)$$

Household Consumption Demand

$$QDFood_{c,cty} = \overline{qDFood0}_{c,cty} \cdot \left(\frac{\overline{gdpPc}_{cty}}{\overline{gdpPc0}_{cty}} \right)^{elaInc_{c,cty}} \\ \cdot \prod_{c2} \left(\frac{(1 - \overline{cse}_{c2,cty}) \cdot PC_{c2,cty}}{(1 - \overline{cse0}_{c2,cty}) \cdot \overline{pc0}_{c2,cty}} \right)^{elaP_{c,c2,cty}} \cdot \frac{\overline{pop}_{cty}}{\overline{pop0}_{cty}} \quad (QDFoodEq)$$

Feed Demand

$$QDFed_{c_fed,cty} = \sum_{bas} \sum_{a_liv} QFeed_{a_liv,c_fed,cty} \quad (QDFedEq)$$

Intermediate Input Demand for Processing

$$QDInt_{c_int,cty} = \sum_{a_pro} QInt_{a_pro,c_int,bas,cty} \quad (QDIntEq)$$

Biofuel Demand

$$QDBio_{c,cty} = \overline{qdBio0}_{c,cty} \cdot \prod_{c2} \left(\frac{(1 - \overline{cse}_{c2,cty}) \cdot PC_{c2,cty}}{(1 - \overline{cse0}_{c2,cty}) \cdot \overline{pc0}_{c2,cty}} \right)^{elaP_{c,c2,cty}} \quad (QDBioEq)$$

Seed Demand

$$QDSeed_{c,cty} = \overline{qdSeed0}_{c,cty} \cdot \frac{\overline{pop}_{cty}}{\overline{pop0}_{cty}} \cdot \prod_{c2} \left(\frac{(1 - \overline{cse}_{c2,cty}) \cdot PC_{c2,cty}}{(1 - \overline{cse0}_{c2,cty}) \cdot \overline{pc0}_{c2,cty}} \right)^{elaP_{c,c2,cty}} \quad (QDSeedEq)$$

Tourist Consumption Demand

$$QDTour_{c,cty} = \overline{qdTour0}_{c,cty} \cdot \prod_{c2} \left(\frac{(1 - \overline{cse}_{c2,cty}) \cdot PC_{c2,cty}}{(1 - \overline{cse0}_{c2,cty}) \cdot \overline{pc0}_{c2,cty}} \right)^{elaP_{c,c2,cty}} \quad (QDTourEq)$$

Loss and Waste

$$QDLoss_{c,cty} = QD_{c,cty} \cdot \overline{lossRate}_{c,cty} \quad (QDLossEq)$$

$$QDWaste_{c,cty} = QDFood_{c,cty} \cdot \overline{wasteRate}_{c,cty} \quad (QDWasteEq)$$

Other Demand

$$QDOth_{c,cty} = \overline{qdOth0}_{c,cty} \cdot \prod_{c2} \left(\frac{(1 - \overline{cse}_{c2,cty}) \cdot PC_{c2,cty}}{(1 - \overline{cse0}_{c2,cty}) \cdot \overline{pc0}_{c2,cty}} \right)^{elaP_{c,c2,cty}} \quad (QDOthEq)$$

4.2 Production Block

The production block describes the equations determining the production quantities of each food commodity, the input of intermediate goods and production factors, and the producer prices. The production functions are represented by Leontief-type functions, where each good is produced through the combination of specific input factors.

4.2.1 Annual Driver Calculations

Crop yields, livestock system shares, and feed input coefficients are calculated before each annual or sub-step equilibrium solve. They are therefore time-varying parameters rather than MCP equilibrium variables:

$$\begin{aligned}
\overline{yld}_{a_cro,bas,cty,wat} &= \overline{yld0}_{a_cro,bas,cty,wat} \cdot \overline{yldTech}_{a_cro,bas,cty,wat} \\
&\quad \cdot \overline{yldChg}_{a_cro,bas,cty,wat} \cdot \overline{yldPlum}_{a_cro,bas,cty}, \\
\overline{sysShr}_{a_liv,bas,cty,cz,LG} &= \frac{1}{1 + \delta_{a_liv,bas,cty,cz,LG}^{sys} \cdot \overline{gdpPc}_{cty}^{-\rho_{a_liv,LG}^{sys}}}, \\
\overline{sysShr}_{a_liv,bas,cty,cz,MX} &= 1 - \overline{sysShr}_{a_liv,bas,cty,cz,LG}, \\
\overline{feedCoef}_{a_liv,c_fed,bas,cty,cz,sys} &= \overline{feedReq}_{a_liv,bas,cty,cz,sys} \cdot \overline{feedConcShr}_{a_liv,bas,cty,cz,sys} \\
&\quad \cdot \overline{feedGrainShr}_{a_liv,c_fed,bas,cty,cz,sys}.
\end{aligned}$$

The optional $\overline{yldPlum}$ term is one when AIM-PLUM feedback is not used.

Domestic Production

$$QS_{a,cty} = \sum_{bas} QSBas_{a,bas,cty} \quad (QSAggEq)$$

4.2.2 Crop Production

Primary crops are differentiated by production technologies (irrigated and rainfed). Yield improvements due to technological progress and climate impacts are considered for crops.

Crop Production by Technology

$$QSWat_{a_cro,bas,cty,wat} = QSBas_{a_cro,bas,cty} \cdot watShr_{a_cro,bas,cty,wat} \quad (QSWatEq)$$

Producer Price of Crops

$$\begin{aligned}
PP_{a_cro,cty} &= \left(PL_{a_cro,bas,cty} \cdot \sum_{wat} \frac{watShr_{a_cro,bas,cty,wat}}{\overline{yld}_{a_cro,bas,cty,wat} \cdot \overline{croInt}_{a_cro,bas,cty,wat}} \right. \\
&\quad \left. + COBas_{a_cro,bas,cty} \right) / (1 + MuBasMin_{a_cro,bas,cty}) \quad (CroZeroProfitEq)
\end{aligned}$$

Land Input for Crop Production

$$QL_{a_cro,bas,cty} = QSBas_{a_cro,bas,cty} \cdot \sum_{wat} \frac{watShr_{a_cro,bas,cty,wat}}{\overline{yld}_{a_cro,bas,cty,wat} \cdot \overline{croInt}_{a_cro,bas,cty,wat}} \quad (CroLandEq)$$

Other Production Factor Price (Crops)

$$COBas_{a_cro,bas,cty} = \overline{coBas0}_{a_cro,bas,cty} \cdot \left(\frac{QSBas_{a_cro,bas,cty}}{qsBasPrev_{a_cro,bas,cty}} \right)^{elaCO_{a_cro,cty}} \quad (COCroEq)$$

4.2.3 Livestock Production

The livestock production block describes equations for determining livestock output, production systems, feed requirements, and associated prices. Livestock products are distinguished by climate zones and production systems (pasture-based or mixed). Feed use and production efficiency depend on feed composition, feed requirements, and animal productivity.

Livestock Production by System

$$QSSys_{a_liv,bas,cty,cz,sys} = QSBas_{a_liv,bas,cty} \cdot czShr_{a_liv,bas,cty,cz} \cdot \overline{sysShr}_{a_liv,bas,cty,cz,sys} \quad (QSSysEq)$$

Producer Price of Livestock

$$\begin{aligned} PP_{a_liv,cty} = & \left(PL_{a_liv,bas,cty} / YLiv_{a_liv,bas,cty} \right. \\ & + feedCostCal_{a_liv,cty} \cdot \sum_{c_fed} (1 - \overline{cse}_{c_fed,cty}) \cdot PC_{c_fed,cty} \\ & \cdot \sum_{cz,sys} \left(czShr_{a_liv,bas,cty,cz} \cdot \overline{sysShr}_{a_liv,bas,cty,cz,sys} \cdot \overline{feedCoef}_{a_liv,c_fed,bas,cty,cz,sys} \right) \\ & \left. + COBas_{a_liv,bas,cty} \right) / (1 + MuBasMin_{a_liv,bas,cty}) \end{aligned} \quad (LivZeroProfitEq)$$

Grass Input Calculation

$$\begin{aligned} QGrass_{a_liv,bas,cty} = & \frac{QSBas_{a_liv,bas,cty}}{\overline{animalProdRate}_{a_liv,cty}} \\ & \cdot \sum_{cz,sys} \left(czShr_{a_liv,bas,cty,cz} \cdot \overline{sysShr}_{a_liv,bas,cty,cz,sys} \right. \\ & \cdot \overline{feedReq}_{a_liv,bas,cty,cz,sys} \cdot (1 - \overline{feedConcShr}_{a_liv,bas,cty,cz,sys}) \left. \right) \end{aligned} \quad (QGrassEq)$$

Available Grassland Area

$$QGrass_{a_liv,bas,cty} = QL_{a_liv,bas,cty} \cdot \overline{yGrass}_{bas,cty} \quad (GrassBalEq)$$

Livestock Yield

$$YLiv_{a_liv,bas,cty} = \frac{QSBas_{a_liv,bas,cty}}{QL_{a_liv,bas,cty}} \quad (YLivEq)$$

Feed Input Calculation

$$QFeed_{a_{liv},c_{fed},bas,cty} = \frac{QSBas_{a_{liv},bas,cty}}{\overline{animalProdRate}_{a_{liv},cty}} \cdot \sum_{\substack{cz,sys}} (\overline{czShr}_{a_{liv},bas,cty,cz} \cdot \overline{sysShr}_{a_{liv},bas,cty,cz,sys}) \cdot \overline{feedCoef}_{a_{liv},c_{fed},bas,cty,cz,sys} \quad (QFeedEq)$$

Other Production Factor Price (Livestock)

$$COBas_{a_{liv},bas,cty} = \overline{coBas0}_{a_{liv},bas,cty} \cdot \left(\frac{QSBas_{a_{liv},bas,cty}}{\overline{qsBasPrev}_{a_{liv},bas,cty}} \right)^{elaCO_{a_{liv},cty}} \quad (COLivEq)$$

Livestock Production Quantity

$$QSBas_{a_{liv},bas,cty} = \frac{\overline{animalWt}_{a_{liv},cty} \cdot (NAnimal_{a_{liv},bas,cty} - \overline{nAnimalRes}_{a_{liv},bas,cty})}{\overline{animalProdRate}_{a_{liv},cty}} \quad (NAnimalEq)$$

4.2.4 Processed Goods production

This section describes the production equations for processed agricultural products. Processed products are produced using intermediate inputs and other production factors.

Producer Price of Processed Products

$$PP_{a_{pro},cty} = \overline{intCostCal}_{a_{pro},cty} \cdot \sum_{c_{int}} \left((1 - \overline{cse}_{c_{int},cty}) \cdot PC_{c_{int},cty} \cdot \overline{intCoef}_{a_{pro},c_{int},cty} \right) + COCty_{a_{pro},cty} \quad (ProZeroProfitEq)$$

Intermediate Input Demand for Processed Products

$$QInt_{a_{pro},c_{int},cty} = QS_{a_{pro},cty} \cdot \overline{intCoef}_{a_{pro},c_{int},cty} \quad (QIntEq)$$

Other Production Factor Price (Processed Products)

$$COCty_{a_{pro},cty} = \overline{coCty0}_{a_{pro},cty} \cdot \left(\frac{QS_{a_{pro},cty}}{\overline{qsPrev}_{a_{pro},cty}} \right)^{elaCO_{a_{pro},cty}} \quad (COProEq)$$

4.2.5 Energy Crop Production

Energy crop production is determined from regional totals provided exogenously by the AIM-Hub model [3].

Regional Energy Crop Supply Balance

$$\overline{qdEne}_{reg} = \sum_{bas,cty \in reg} QSBa_{secr,bas,cty} \quad (\text{EneBalEq})$$

Land Input for Energy Crops

$$QL_{ecr,bas,cty} = \frac{QSBa_{secr,bas,cty}}{\overline{yld}_{ecr,bas,cty,Rainfed}} \quad (\text{EneLandEq})$$

Land Price for Energy Crops

$$PL_{ecr,bas,cty} = PLEne_{reg} \quad (\text{PLEneEq})$$

4.3 Price Block

This subsection describes the equations in the price block. The price block represents the relationships among various prices in the model. Prices are linked through fixed coefficients so that the price used in the demand and supply functions is consistently determined.

Consumer and Producer Prices

$$(1 + \overline{mmj}_{a,cty}) \cdot PP_{a,cty} + CEmis_{a,cty} + CAbt_{a,cty} = PC_{a,cty} \quad (\text{PCNtrdEq})$$

Consumer and Border Prices

$$PC_{c,cty} = (1 + \overline{pse}_{c,cty}) \cdot PB_{c,cty} \cdot (1 + pbRes_{c,cty}) \quad (\text{PCTrdEq})$$

Import Price

$$PM_{c,cty} = PW_c \cdot exr_{cty} \cdot (1 + \overline{tm}_{c,cty} + MuImpMin_{c,cty}) \cdot (1 + \overline{mmm}_{c,cty}) \cdot (1 + pmRes_{c,cty}) \quad (\text{PMEq})$$

Export Price

$$PE_{c,cty} \cdot (1 + \overline{te}_{c,cty}) \cdot (1 + \overline{mme}_{c,cty}) = PW_c \cdot (1 + peRes_{c,cty}) \cdot exr_{cty} \quad (\text{PEEq})$$

4.4 Land Use Block

The land-use block allocates available land among different uses through a Logit function. Non-pasture vegetation is allocated among cropland, forest, other natural land, and afforested land. Protected areas are imposed as lower bounds for the corresponding land classes, and the agro-ecological suitability constraint applies to cropland. Urban land is fixed exogenously. In the default configuration, the LUH2-consistent base-year urban area is held constant over future years; as an optional extension, SSP-specific urban projections from the SEDAC urban dataset can be used to update urban land over time.

Vegetation Area

$$QLVeg_{bas,cty} = \overline{qITot}_{bas,cty} - \overline{qLOth}_{bas,cty} - \overline{qLUrb}_{bas,cty} \quad (QLVegBalEq)$$

Pasture Area Allocation

$$QLPas_{bas,cty} = QLVeg_{bas,cty} \cdot \frac{\delta_{bas,cty}^{pas} \cdot PLPas_{bas,cty}^{\rho_{bas,cty}^{pas}}}{\delta_{bas,cty}^{pas} \cdot PLPas_{bas,cty}^{\rho_{bas,cty}^{pas}} + \delta_{bas,cty}^{pasn} \cdot PLNpas_{bas,cty}^{\rho_{bas,cty}^{pasn}}} \quad (PasLogitEq)$$

Grazed and Ungrazed Pasture Allocation

$$QLPasGrz_{bas,cty} = QLPas_{bas,cty} \cdot \frac{\delta_{bas,cty}^{pasgrz} \cdot PLPasGrz_{bas,cty}^{\rho_{bas,cty}^{pasgrz}}}{\delta_{bas,cty}^{pasgrz} \cdot PLPasGrz_{bas,cty}^{\rho_{bas,cty}^{pasgrz}} + \delta_{bas,cty}^{pasung} \cdot PLPasUng_{bas,cty}^{\rho_{bas,cty}^{pasung}}} \quad (PasGrzLogitEq)$$

$$QLPasUng_{bas,cty} = QLPas_{bas,cty} \cdot \frac{\delta_{bas,cty}^{pasung} \cdot PLPasUng_{bas,cty}^{\rho_{bas,cty}^{pasung}}}{\delta_{bas,cty}^{pasgrz} \cdot PLPasGrz_{bas,cty}^{\rho_{bas,cty}^{pasgrz}} + \delta_{bas,cty}^{pasung} \cdot PLPasUng_{bas,cty}^{\rho_{bas,cty}^{pasung}}} \quad (PasUngLogitEq)$$

Non-pasture Grassland Area

$$QLNpas_{bas,cty} = QLVeg_{bas,cty} \cdot \frac{\delta_{bas,cty}^{pasn} \cdot PLNpas_{bas,cty}^{\rho_{bas,cty}^{pasn}}}{\delta_{bas,cty}^{pas} \cdot PLPas_{bas,cty}^{\rho_{bas,cty}^{pas}} + \delta_{bas,cty}^{pasn} \cdot PLNpas_{bas,cty}^{\rho_{bas,cty}^{pasn}}} \quad (NpasLogitEq)$$

Cropland Area

$$QLCro_{bas,cty} = QLNpas_{bas,cty}$$

$$\frac{\delta_{bas,cty}^{cro} PLCro_{bas,cty}^{\rho_{bas,cty}^{cro}}}{\delta_{bas,cty}^{cro} PLCro_{bas,cty}^{\rho_{bas,cty}^{cro}} + \delta_{bas,cty}^{frs} PLFrS_{bas,cty}^{\rho_{bas,cty}^{frs}} + \delta_{bas,cty}^{gra} PLNat_{bas,cty}^{\rho_{bas,cty}^{gra}} + \delta_{bas,cty}^{afr} PLAfr_{bas,cty}^{\rho_{bas,cty}^{afr}}} \quad (CroLogitEq)$$

Forest Area

$$QLFrS_{bas,cty} = QLNpas_{bas,cty}$$

$$\frac{\delta_{bas,cty}^{frs} PLFrS_{bas,cty}^{\rho_{bas,cty}^{frs}}}{\delta_{bas,cty}^{cro} PLCro_{bas,cty}^{\rho_{bas,cty}^{cro}} + \delta_{bas,cty}^{frs} PLFrS_{bas,cty}^{\rho_{bas,cty}^{frs}} + \delta_{bas,cty}^{gra} PLNat_{bas,cty}^{\rho_{bas,cty}^{gra}} + \delta_{bas,cty}^{afr} PLAfr_{bas,cty}^{\rho_{bas,cty}^{afr}}} \quad (FrSLogitEq)$$

Primary and Secondary Forest Allocation

$$QLPrimf_{bas,cty} = QLFrs_{bas,cty} \cdot \frac{\delta_{bas,cty}^{primf} PLPrimf_{bas,cty}^{\rho_{bas,cty}^{primf}}}{\delta_{bas,cty}^{primf} PLPrimf_{bas,cty}^{\rho_{bas,cty}^{primf}} + \delta_{bas,cty}^{secdf} PLSecdf_{bas,cty}^{\rho_{bas,cty}^{secdf}}} \quad (PrimfLogitEq)$$

$$QLSecdf_{bas,cty} = QLFrs_{bas,cty} \cdot \frac{\delta_{bas,cty}^{secdf} PLSecdf_{bas,cty}^{\rho_{bas,cty}^{secdf}}}{\delta_{bas,cty}^{primf} PLPrimf_{bas,cty}^{\rho_{bas,cty}^{primf}} + \delta_{bas,cty}^{secdf} PLSecdf_{bas,cty}^{\rho_{bas,cty}^{secdf}}} \quad (SecdfLogitEq)$$

Other Natural Land

$$QLNat_{bas,cty} = QLNpas_{bas,cty}$$

$$\frac{\delta_{bas,cty}^{gra} PLNat_{bas,cty}^{\rho_{bas,cty}^{gra}}}{\delta_{bas,cty}^{cro} PLCro_{bas,cty}^{\rho_{bas,cty}^{cro}} + \delta_{bas,cty}^{frs} PLFrS_{bas,cty}^{\rho_{bas,cty}^{frs}} + \delta_{bas,cty}^{gra} PLNat_{bas,cty}^{\rho_{bas,cty}^{gra}} + \delta_{bas,cty}^{afr} PLAfr_{bas,cty}^{\rho_{bas,cty}^{afr}}} \quad (NatLogitEq)$$

Afforested Land

$$QLAfr_{bas,cty} = QLNpas_{bas,cty} \cdot \frac{\delta_{bas,cty}^{afr} PLAfr_{bas,cty}^{\rho_{bas,cty}^{afr}}}{\delta_{bas,cty}^{cro} PLCro_{bas,cty}^{\rho_{bas,cty}^{cro}} + \delta_{bas,cty}^{frs} PLFr_{bas,cty}^{\rho_{bas,cty}^{frs}} + \delta_{bas,cty}^{gra} PLNat_{bas,cty}^{\rho_{bas,cty}^{gra}} + \delta_{bas,cty}^{afr} PLAfr_{bas,cty}^{\rho_{bas,cty}^{afr}}} \quad (AfrLogitEq)$$

Harvested Cropland Area

$$QLCroCul_{bas,cty} = QLCro_{bas,cty} \cdot \frac{\delta_{bas,cty}^{croclt} PLCroCul_{bas,cty}^{\rho_{bas,cty}^{croclt}}}{\delta_{bas,cty}^{croclt} PLCroCul_{bas,cty}^{\rho_{bas,cty}^{croclt}} + \delta_{bas,cty}^{crofal} PLCroFal_{bas,cty}^{\rho_{bas,cty}^{crofal}}} \quad (CroCulLogitEq)$$

Fallow Cropland Area

$$QLCroFal_{bas,cty} = QLCro_{bas,cty} \cdot \frac{\delta_{bas,cty}^{crofal} PLCroFal_{bas,cty}^{\rho_{bas,cty}^{crofal}}}{\delta_{bas,cty}^{croclt} PLCroCul_{bas,cty}^{\rho_{bas,cty}^{croclt}} + \delta_{bas,cty}^{crofal} PLCroFal_{bas,cty}^{\rho_{bas,cty}^{crofal}}} \quad (CroFalLogitEq)$$

Cropland Allocation

$$QLa_{cro,bas,cty} = QLCroCul_{bas,cty} \cdot \frac{\delta_{a_{cro,bas,cty}} \cdot PL_{a_{cro,bas,cty}}^{\rho_{a_{cro,bas,cty}}}}{\sum_{a_{cro}} \delta_{a_{cro,bas,cty}} \cdot PL_{a_{cro,bas,cty}}^{\rho_{a_{cro,bas,cty}}}} \quad (CroDistEq)$$

Pasture Allocation

$$QLa_{liv,bas,cty} = QLPasGrz_{bas,cty} \cdot \frac{\delta_{a_{liv,bas,cty}} \cdot PL_{a_{liv,bas,cty}}^{\rho_{a_{liv,bas,cty}}}}{\sum_{a_{liv}} \delta_{a_{liv,bas,cty}} \cdot PL_{a_{liv,bas,cty}}^{\rho_{a_{liv,bas,cty}}}} \quad (LivDistEq)$$

Land Prices The price drivers for other natural land and afforestation include their land-carbon value before the associated complementarity wedges are applied.

$$PLCroCul_{bas,cty} \cdot QLCroCul_{bas,cty} = \sum_{a_{cro}} PL_{a_{cro,bas,cty}} \cdot QLa_{cro,bas,cty} \quad (PLCroCulEq)$$

$$PLCro_{bas,cty} \cdot QLCro_{bas,cty} = (PLCroCul_{bas,cty} \cdot QLCroCul_{bas,cty} + PLCroFal_{bas,cty} \cdot QLCroFal_{bas,cty}) / (1 + MuCroCap_{bas,cty}) \quad (PLCroEq)$$

$$\begin{aligned}
 PLNpas_{bas,cty}^{\rho_{bas,cty}^{pasn}} \cdot \overline{qlNpas0}_{bas,cty} &= \overline{plNpas0}_{bas,cty}^{\rho_{bas,cty}^{pasn}} \\
 &\cdot \left(\delta^{cro} PLCro^{\rho^{cro}} + \delta^{frs} PLFrS^{\rho^{frs}} + \delta^{gra} PLNat^{\rho^{gra}} + \delta^{afr} PLAfR^{\rho^{afr}} \right)_{bas,cty} \quad (PLNpasEq)
 \end{aligned}$$

$$PLPasGrz_{bas,cty} \cdot QLPasGrz_{bas,cty} = \sum_{a_{liv}} PL_{a_{liv},bas,cty} \cdot QL_{a_{liv},bas,cty} \quad (PLPasGrzEq)$$

$$\begin{aligned}
 PLPas_{bas,cty} \cdot QLPas_{bas,cty} &= \\
 & (PLPasGrz_{bas,cty} \cdot QLPasGrz_{bas,cty} + PLPasUng_{bas,cty} \cdot QLPasUng_{bas,cty}) \\
 & \cdot (1 + MuPasProt_{bas,cty}) \quad (PLPasEq)
 \end{aligned}$$

$$\begin{aligned}
 PLFrS_{bas,cty}^{\rho_{bas,cty}^{frs}} \cdot \overline{qlFrS0}_{bas,cty} &= \overline{plFrS0}_{bas,cty}^{\rho_{bas,cty}^{frs}} \\
 & \cdot \left(\delta^{primf} PLPrimf^{\rho^{primf}} + \delta^{secd f} PLSecdf^{\rho^{secd f}} \right)_{bas,cty} \quad (PLFrSEq)
 \end{aligned}$$

$$\begin{aligned}
 PLPrimf_{bas,cty} &= \left[\overline{plPrimf0}_{bas,cty} + \max \left(0, \overline{carbonStock}_{bas,cty} \cdot \frac{44}{12} \right. \right. \\
 & \quad \left. \left. \cdot \overline{pCarLand}_{bas,cty} \cdot dcf frs_{cty} - afrCost_{cty} \right) \right] \\
 & \cdot \frac{1 + MuPrimfProt_{bas,cty}}{1 + MuPrimfCap_{bas,cty}} \quad (PLPrimfEq)
 \end{aligned}$$

$$\begin{aligned}
 PLSecdf_{bas,cty} &= \left[\overline{plSecdf0}_{bas,cty} + \max \left(0, \overline{carbonStock}_{bas,cty} \cdot \frac{44}{12} \right. \right. \\
 & \quad \left. \left. \cdot \overline{pCarLand}_{bas,cty} \cdot dcf frs_{cty} - afrCost_{cty} \right) \right] \\
 & \cdot \frac{1 + MuSecdfProt_{bas,cty}}{1 + MuSecdfCap_{bas,cty}} \quad (PLSecdfEq)
 \end{aligned}$$

$$PLNat_{bas,cty} = \overline{plNatNet}_{bas,cty} \cdot (1 + MuNatProt_{bas,cty}) \quad (PLNatEq)$$

Afforestation Price and Annual Limit

$$PLAfr_{bas,cty} = \frac{\overline{plAfrRaw}_{bas,cty}}{1 + \overline{MuAfrCap}_{bas,cty}} \quad (PLAfrEq)$$

$$QLAfr_{bas,cty} \leq QLAfr_{bas,cty}^{t-1} + \overline{afrPlantMax} \cdot \overline{qlNpas0}_{bas,cty} \\ \perp \quad \overline{MuAfrCap}_{bas,cty} \geq 0 \quad (AfrCapEq)$$

Forest Expansion Constraints

$$QLPrimf_{bas,cty} \leq QLPrimf_{bas,cty}^{t-1} \cdot (1 + \overline{primfExpRate}_{bas,cty}) \\ \perp \quad \overline{MuPrimfCap}_{bas,cty} \geq 0 \quad (PrimfCapEq)$$

$$QLSecdf_{bas,cty} \leq QLSecdf_{bas,cty}^{t-1} \cdot (1 + \overline{secdfExpRate}_{bas,cty}) \\ \perp \quad \overline{MuSecdfCap}_{bas,cty} \geq 0 \quad (SecdfCapEq)$$

Protected-Area Constraints

$$QLPrimf_{bas,cty} \geq \overline{qlProt}_{bas,cty,primf} \quad \perp \quad \overline{MuPrimfProt}_{bas,cty} \geq 0, \quad (PrimfProtEq)$$

$$QLSecdf_{bas,cty} \geq \overline{qlProt}_{bas,cty,secdf} \quad \perp \quad \overline{MuSecdfProt}_{bas,cty} \geq 0, \quad (SecdfProtEq)$$

$$QLNat_{bas,cty} \geq \overline{qlProt}_{bas,cty,gra} \quad \perp \quad \overline{MuNatProt}_{bas,cty} \geq 0, \quad (NatProtEq)$$

$$QLPas_{bas,cty} \geq \sum_{lu_pa_pas} \overline{qlProt}_{bas,cty,lu_pa_pas} \quad \perp \quad \overline{MuPasProt}_{bas,cty} \geq 0. \quad (PasProtEq)$$

Cropland Constraint

$$\overline{qlCroCap}_{bas,cty} \geq QLCro_{bas,cty} \quad \perp \quad \overline{MuCroCap}_{bas,cty} \geq 0 \quad (CroCapEq)$$

4.5 Market Block

This subsection describes the equations in the market block, where consistency between food demand and supply is ensured. Trade of food commodities is represented as a complementarity problem. When the import price is higher than the border price, the import quantity becomes zero. Once the import price decreases to equal the border price, imports begin. Similarly, when the border price is higher than the export price, exports become zero, and once the export price rises to match the border price, exports take place.

Domestic Trade Balance

$$NT_{c,cty} = QS_{c,cty} - QD_{c,cty} - \overline{qStock}_{c,cty} \quad (\text{ComBalEq})$$

$$NT_{c,cty} = QE_{c,cty} - QM_{c,cty} \quad (\text{NTEq})$$

International Trade Balance

$$\sum_{cty} NT_{c,cty} - \overline{qRes}_c = 0 \quad (\text{WldBalEq})$$

Import and Export Conditions

$$PM_{c,cty} \geq PB_{c,cty} \quad \perp \quad QM_{c,cty} \geq 0 \quad (\text{ImpParityEq})$$

$$PB_{c,cty} \geq PE_{c,cty} \quad \perp \quad QE_{c,cty} \geq 0 \quad (\text{ExpParityEq})$$

4.6 GHG Emission Block

This subsection describes the equations in the GHG emission block. GHG emissions are calculated based on the production quantities of each food commodity. When carbon prices are considered, the amount of mitigation technology adoption and its cost are computed from the MAC (Marginal Abatement Cost) curve. Emissions from the land-use sector are calculated according to each land area.

GHG Emissions

$$GHG_{a,src,gas,cty} = QS_{a,cty} \cdot \overline{ef}_{a,src,gas,cty} \cdot (1 - RRed_{a,src,gas,cty}) \quad (\text{GHGAgEq})$$

Land-Use Emissions

$$\begin{aligned} GHGLU_{src,gas,cty} = & \sum_{bas} \left(\overline{efLUCro}_{src,gas,cty} \cdot QLCro_{bas,cty} \right. \\ & + \overline{efLUFrs}_{src,gas,cty} \cdot QLFr_{bas,cty} \\ & - \overline{carbonStock}_{bas,cty} \cdot \frac{44}{12} \cdot \min\left\{0, QLFr_{bas,cty} - QLFr_{bas,cty}^{t-1}\right\} \\ & \left. + \overline{efLUPas}_{src,gas,cty} \cdot QLPas_{bas,cty} + \overline{efLUNat}_{src,gas,cty} \cdot QLNat_{bas,cty} \right) \quad (\text{GHGLUEq}) \end{aligned}$$

$$GHGLU_{afr,CO_2,cty} = - \sum_{bas} \sum_{age} qLAfrAge_{bas,cty,age} \cdot \overline{carbonSeq}_{bas,cty,age} \quad (\text{AfrGHGEq})$$

Emission Reduction Rate

$$RRed_{a,src,gas,cty} = \overline{mrp}_{src,gas,cty} \cdot \left(1 - \left[1 / (1 + \overline{pCar}_{cty}) \right]^{\sigma_{src,gas,cty}} \right) \quad (\text{RRedEq})$$

Abatement Cost

$$CAbt_{a,cty} = \sum_{src,gas} \overline{ef}_{a,src,gas,cty} \cdot \left[\frac{\sigma_{src,gas,cty}}{(\sigma_{src,gas,cty} - 1)} \cdot \overline{mrp}_{src,gas,cty} \cdot \left(1 - \left(1 - RRed_{a,src,gas,cty} / \overline{mrp}_{src,gas,cty} \right)^{(1-1/\sigma_{src,gas,cty})} \right) - RRed_{a,src,gas,cty} \right] \quad (CAbtEq)$$

Carbon Cost on Remaining Emissions

$$CEmis_{a,cty} = \sum_{src,gas} \overline{pCar}_{cty} \cdot \overline{ef}_{a,src,gas,cty} \cdot (1 - RRed_{a,src,gas,cty}) \quad (CEmisEq)$$

4.7 Complementarity Conditions

This subsection describes the complementarity conditions introduced to avoid numerical errors in the Logit function when production quantities in specific production units become zero. If the production of a unit becomes zero, its land-use area would also become zero, leading to instability. Therefore, a minimum positive value is imposed to ensure that production quantities do not fall to zero.

Minimum Production Share for Trade

$$\frac{QS_{c,cty}}{QM_{c,cty}} \geq 0.01 \quad \perp \quad MuImpMin_{c,cty} \geq 0 \quad (ImpMinEq)$$

Minimum Production Share for Supply

$$\frac{QSBas_{a,bas,cty}}{QS_{a,cty}} \geq 0.01 \quad \perp \quad MuBasMin_{a,bas,cty} \geq 0 \quad (QSBasMinEq)$$

5 Data and Preprocessing

This section explains the calibration and processing of data used in the model. Exogenous variables and parameters were estimated using multiple datasets, adjusted to obtain equilibrium in the base year, and parameters were estimated by regression analysis.

5.1 Data Sources

Table 6 lists the datasets used in the model.

Table 6: List of data sources

Data Source	Dataset Name	URL
FAOSTAT [9]	Food Balances	https://www.fao.org/faostat/en/#data/FBS
FAOSTAT [9]	Production	https://www.fao.org/faostat/en/#data/QCL
FAOSTAT [9]	Trade	https://www.fao.org/faostat/en/#data/TCL
FAOSTAT [9]	Producer Prices	https://www.fao.org/faostat/en/#data/PP
FAOSTAT [9]	Land Use	https://www.fao.org/faostat/en/#data/RL
FAOSTAT [9]	Emissions	https://www.fao.org/faostat/en/#data/GCE
FAO Land and Water Division [2]	Major Hydrological Basins of the World	https://data.apps.fao.org/catalog//iso/7707086d-af3c-41cc-8aa5-323d8609b2d1
FAO and IIASA [10]	GAEZ v4	https://gaez.fao.org/
IIASA	SSP 3.0	https://data.ece.iiasa.ac.at/ssp/#/about
GTAP [11]	GTAP 10	https://jgea.org/ojs/index.php/jgea/article/view/77
IFPRI [12]	MAPSPAM	https://doi.org/10.7910/DVN/SWPENT
IFPRI [13]	IMPACT	https://github.com/IFPRI/IMPACT
World Bank	World Bank Open Data	https://data.worldbank.org/
EDGAR [14]	EDGAR v8.0	https://edgar.jrc.ec.europa.eu/dataset_ghg80
Herrero et al. (2018) [15]	Livestock production systems	https://doi.org/10.4225/08/5aa068b33fe06
Muhammad et al. (2011) [16]	International Evidence on Food Consumption Patterns	http://199.135.94.241/publications/pub-details/?pubid=47581
T. P. Robinson et al. (2018) [17]	Global distribution of ruminant livestock production systems V5	https://doi.org/10.7910/DVN/WPDSZE

Continued on next page

Table 6: List of data sources (continued)

Data Source	Dataset Name	URL
Harmsen et al. (2019) [18]	CH ₄ and N ₂ O MAC curves	https://doi.org/10.1016/j.envsci.2019.05.013
Hurt et al. (2019) [19]	Land Use Harmonization 2	https://luh.umd.edu/index.shtml
Hanasaki et al. (2008) [20, 21]	H08 rainfed crop yield	https://doi.org/10.5194/hess-12-1007-2008
Hasegawa et al. (2017) [5]	AIM-PLUM spatial land-use outputs	https://doi.org/10.1016/j.scitotenv.2016.12.025
Ito and Inatomi (2012) [22]	VISIT carbon-cycle outputs	https://doi.org/10.1175/JHM-D-10-05034.1
Gao and Pesaresi (2021) [23]	Global 1-km Downscaled Urban Land Extent Projection and Base Year Grids by SSP Scenarios, 2000-2100	https://doi.org/10.7927/1z4r-ez63
Gao and Pesaresi (2021) [24]	Downscaling SSP-consistent Global Spatial Urban Land Projections from 1/8-degree to 1-km Resolution, 2000-2100	https://doi.org/10.1038/s41597-021-01052-0
IUCN and UNEP-WCMC [25]	World Database on Protected Areas (WDPA), Dec 2025 release	https://www.protectedplanet.net

5.2 Calibration and processing of data

5.2.1 Adjustment of demand-side data

First, the adjustment of demand-side data was conducted. In the model, if a country–commodity pair has missing data for the base year 2015, it is assumed that consumption does not occur thereafter. Therefore, for country–commodity pairs without 2015 data in the Food Balance Sheets (FBS), the average value of subsequent years was used as the 2015 consumption level.

$$qdFood_{(c,cty,2015)} = \frac{\sum_{yr} qdFood_{(c,cty,yr)}^{FAO}}{N_{yr}} \quad \text{for } qdFood_{(c,cty,2015)} = \text{NA} \quad (5.1)$$

where $qdFood^{FAO}$ is household consumption data from FAO, and N_{yr} is the number of years for which data exist.

In the FBS data, household consumption includes the amount of food wasted at the household level. To distinguish this, waste rates by seven commodities and seven regions were obtained from FAO (2011) [26]. Using these coefficients, household consumption and food waste were estimated as follows:

$$qdWaste_{(c,cty)} = qdFood_{(c,cty)}^{FAO} \cdot wasteRate_{(c,cty)} \quad (5.2)$$

$$qdFood_{(c,cty)} = qdFood_{(c,cty)}^{FAO} \cdot (1 - wasteRate_{(c,cty)}) \quad (5.3)$$

For model simplification, the commodities used as livestock feed, processing materials, and biofuel inputs were limited. Wheat, rice, maize, other cereals, roots and tubers, sugar crops, pulses, oil crops, and vegetables were used as livestock feed. Wheat, rice, maize, other cereals, and fruits were used as raw materials for alcoholic beverages. Sugar crops were used as inputs for sugars, oil crops for vegetable oils, and maize, sugar crops, and vegetable oils for biofuels. Feed demand, intermediate input demand, and biofuel demand for other commodities were set to zero.

$$qdFed_{(c \notin c_{fed}, cty)} = 0 \quad (5.4)$$

$$qdBio_{(c \notin c_{bio}, cty)} = 0 \quad (5.5)$$

$$qdInt_{(c \notin c_{int}, cty)} = 0 \quad (5.6)$$

As a result of these adjustments, the total domestic demand changed. If the sum of all demand components was less than the total demand from the data, the residual was added as “other demand”:

$$qdOth_{(c,cty)} = qd_{(c,cty)} - (qdFood_{(c,cty)} + qdFed_{(c,cty)} + qdInt_{(c,cty)} + qdSeed_{(c,cty)} + qdTour_{(c,cty)} + qdLoss_{(c,cty)} + qdWaste_{(c,cty)}) \quad (5.7)$$

5.2.2 Adjustment of crop production data

Next, the adjustment of crop production data was conducted. From the FBS data, the production quantities of primary crops for each country were obtained. These were disaggregated into production units and production technologies using the GAEZ data as follows:

$$qsWat_{(a_{cro}, bas, cty, wat)} = qs_{(a_{cro}, cty)}^{FAO} \cdot \frac{qsGaez_{(a_{cro}, bas, cty, wat)}^{GAEZ}}{\sum_{(bas, wat)} qsGaez_{(a_{cro}, bas, cty, wat)}^{GAEZ}} \quad (5.8)$$

where qs^{FAO} denotes production quantity data from FAO, and $qsGaez^{GAEZ}$ represents production data from GAEZ.

From the LUH2 dataset, the cultivated land area for each production unit was obtained. Using the FAOSTAT data on national land use, these cultivated lands were divided into harvested and fallow lands as follows:

$$qlCroCul_{(bas,cty)} = qlCro_{(bas,cty)}^{LUH2} \cdot \frac{qlCroCul_{cty}^{FAO}}{qlCroCul_{cty}^{FAO} + qlCroFal_{cty}^{FAO}} \quad (5.9)$$

$$qlCroFal_{(bas,cty)} = qlCro_{(bas,cty)}^{LUH2} \cdot \frac{qlCroFal_{cty}^{FAO}}{qlCroCul_{cty}^{FAO} + qlCroFal_{cty}^{FAO}} \quad (5.10)$$

where $qlCro^{LUH2}$ denotes cultivated area data from LUH2, $qlCroCul^{FAO}$ is harvested land area data from FAO, and $qlCroFal^{FAO}$ is fallow land area data from FAO.

Furthermore, the harvested land was disaggregated by crop type and production technology using GAEZ data as follows:

$$qlWat_{(a_cro,bas,cty,wat)} = qlCroCul_{(bas,cty)} \cdot \frac{qlGaez_{(a_cro,bas,cty,wat)}^{GAEZ}}{\sum_{(a_cro,wat)} qlGaez_{(a_cro,bas,cty,wat)}^{GAEZ}} \quad (5.11)$$

where $qlWat$ is the cropland area by crop type and production technology, and $qlGaez^{GAEZ}$ is the crop-specific land area from GAEZ.

From the MAPSPAM dataset, harvested and physical area data were obtained for each crop, production unit, and production technology. Using these data, the cropping intensity was estimated as follows:

$$croInt_{(a_cro,bas,cty,wat)} = \frac{mapspam_{(a_cro,bas,cty,wat)}^{Harvested}}{mapspam_{(a_cro,bas,cty,wat)}^{Physical}} \quad (5.12)$$

where $mapspam^{Harvested}$ is the harvested area data from MAPSPAM, and $mapspam^{Physical}$ is the physical area data from MAPSPAM.

Crop yields were calculated using the obtained production quantities, harvested areas, and cropping intensities as follows:

$$yld_{(a_cro,bas,cty,wat)} = \frac{qsWat_{(a_cro,bas,cty,wat)}}{qlWat_{(a_cro,bas,cty,wat)} \cdot croInt_{(a_cro,bas,cty,wat)}} \quad (5.13)$$

Food-crop yields are based on FAOSTAT production and GAEZ spatial allocation. For dedicated energy crops, AIM-ALPHA instead uses the rainfed yield from H08, following the bioenergy-potential assessment of Wu et al. [20, 21, 27].

When feedback from AIM-PLUM is enabled, the crop-yield driver is adjusted by the change in basin-average yield associated with changes in the grid-level spatial allocation of crop production [5]. The adjustment is one when this feedback is not used.

5.2.3 Adjustment of livestock production data

Next, the adjustment of livestock production data was conducted. From FAO data, the numbers of livestock animals and slaughtered (or milked) animals were obtained. Using these, the production rate was calculated. A minimum threshold of 0.3 was set for the production rate; if it fell below this value, it was assumed that a portion of livestock was not involved in production activities.

$$animalProdRate_{(a_liv,cty)} = \frac{nAnimal^{FAO}_{(a_liv,cty)}}{nAnimalStock^{FAO}_{(a_liv,cty)}} \quad (5.14)$$

$$nAnimalRes_{(a_liv,cty)} = nAnimalStock^{FAO}_{(a_liv,cty)} - \frac{nAnimal^{FAO}_{(a_liv,cty)}}{0.3} \quad (5.15)$$

(if $animalProdRate_{(a_liv,cty)} < 0.3$)

where $nAnimal^{FAO}$ is the number of slaughtered (or milked) animals from FAO, and $nAnimalStock^{FAO}$ is the total livestock population.

When non-productive livestock exist, the corresponding unused pasture area was estimated as:

$$qlPasRes_{(a_liv,cty)} = qlPas^{LUH2}_{(a_liv,cty)} \cdot \frac{nAnimalRes_{(a_liv,cty)}}{nAnimalStock^{FAO}_{(a_liv,cty)}} \quad (5.16)$$

where $qlPasRes$ denotes the unused pasture area, and $qlPas^{LUH2}$ is the pasture area from LUH2.

Country-level livestock production data from the FBS were disaggregated into production units, climate zones, and production systems using the dataset from Herrero et al. (2018). Similarly, LUH2 pasture area data were disaggregated by livestock type, climate zone, and production system.

$$qsSys_{(a_liv,bas,cty,cz,sys)} = qs^{FAO}_{(a_liv,cty)} \cdot \frac{livWt^{HERRERO}_{(a_liv,bas,cty,cz,sys)}}{\sum_{(bas,cz,sys)} livWt^{HERRERO}_{(a_liv,bas,cty,cz,sys)}} \quad (5.17)$$

$$qlSys_{(a_liv,bas,cty,cz,sys)} = (qlPas^{LUH2}_{(cty,bas)} - qlPasRes_{(a_liv,cty)}) \cdot grassUseRate_{(bas,cty)} \cdot \frac{pasArea^{HERRERO}_{(a_liv,bas,cty,cz,sys)}}{\sum_{(a_liv,cz,sys)} pasArea^{HERRERO}_{(a_liv,bas,cty,cz,sys)}} \quad (5.18)$$

where $livWt^{HERRERO}$ denotes production data and $pasArea^{HERRERO}$ represents pasture use data from Herrero et al. (2018). The grass utilization rate is calculated from estimated grass demand and maximum available supply, and capped in calibration to avoid overuse.

The grazed and ungrazed pasture areas are then computed as:

$$qlPasGrz(bas,cty) = \sum_{(a_liv,cz,sys)} qlSys_{(a_liv,bas,cty,cz,sys)} \quad (5.19)$$

$$qlPasUng(bas,cty) = qlPas_{(cty,bas)}^{LUH2} - qlPasGrz(bas,cty) \quad (5.20)$$

Next, the feed requirements and concentrate feed share were estimated as follows:

$$feedReq_{(a_liv,bas,cty,cz,sys)} = \frac{feedQty_{(a_liv,bas,cty,cz,sys)}^{HERRERO} + pasArea_{(a_liv,bas,cty,cz,sys)}^{HERRERO}}{livWt_{(a_liv,bas,cty,cz,sys)}^{HERRERO}} \quad (5.21)$$

$$feedConcShr_{(a_liv,bas,cty,cz,sys)} = \frac{feedQty_{(a_liv,bas,cty,cz,sys)}^{HERRERO}}{feedQty_{(a_liv,bas,cty,cz,sys)}^{HERRERO} + pasArea_{(a_liv,bas,cty,cz,sys)}^{HERRERO}} \quad (5.22)$$

where $feedQty^{HERRERO}$ is the amount of concentrate feed from Herrero et al. (2018).

Using these values and the FBS feed demand data, the quantities of pasture and concentrate feed use were estimated. The amount of concentrate feed input was adjusted with a correction coefficient to ensure consistency with the FBS data. Consequently, the feed requirements and concentrate feed share were also adjusted. If the estimated pasture use exceeded the available supply, it was replaced with the available amount.

$$qFeedSys_{(a_liv,c_fed,bas,cty,cz,sys)}^{temp} = \frac{qsSys_{(a_liv,bas,cty,cz,sys)}}{animalProdRate_{(a_liv,cty)}} \cdot feedReq_{(a_liv,bas,cty,cz,sys)} \cdot feedConcShr_{(a_liv,bas,cty,cz,sys)} \cdot \frac{qdFed_{(c_fed,cty)}^{FAO}}{\sum_{c_fed} qdFed_{(c_fed,cty)}^{FAO}} \quad (5.23)$$

$$feedScale_{(c_fed,cty)} = \frac{qdFed_{(c_fed,cty)}^{FAO}}{\sum_{(a_liv,bas,cz,sys)} qFeedSys_{(a_liv,c_fed,bas,cty,cz,sys)}^{temp}} \quad (5.24)$$

$$qFeedSys_{(a_liv,c_fed,bas,cty,cz,sys)} = qFeedSys_{(a_liv,c_fed,bas,cty,cz,sys)}^{temp} \cdot feedScale_{(c_fed,cty)} \quad (5.25)$$

$$qGrassSys(a_{liv}, bas, cty, cz, sys) = \frac{qsSys(a_{liv}, bas, cty, cz, sys)}{animalProdRate(a_{liv}, cty)} \cdot feedReq(a_{liv}, bas, cty, cz, sys) \cdot (1 - feedConcShr(a_{liv}, bas, cty, cz, sys)) \quad (5.26)$$

If the estimated pasture demand exceeded the available supply, it was replaced with the available amount:

$$qGrassSys(a_{liv}, bas, cty, cz, sys) = qlSys(a_{liv}, bas, cty, cz, sys) \cdot yGrass(bas, cty) \quad \text{if } qlSys(a_{liv}, bas, cty, cz, sys) \cdot yGrass(bas, cty) < qGrassSys(a_{liv}, bas, cty, cz, sys) \quad (5.27)$$

where:

- $qFeedSys^{temp}$: estimated feed demand by production system,
- $qFed^{FAO}$: feed demand data from FAO,
- $feedScale$: adjustment coefficient for consistency with FAO feed demand,
- $qFeedSys$: adjusted feed demand consistent with FAO data,
- $qGrassSys$: pasture demand by production system.

5.2.4 Adjustment of food trade

Next, the adjustment of food trade data was conducted. In order to express trade through complementary conditions, the net import and net export quantities were derived as follows:

$$qm_{(c, cty)} = qm_{(c, cty)}^{FAO} - qe_{(c, cty)}^{FAO} \quad (\text{if } qm_{(c, cty)}^{FAO} > qe_{(c, cty)}^{FAO}) \quad (5.28)$$

$$qe_{(c, cty)} = qe_{(c, cty)}^{FAO} - qm_{(c, cty)}^{FAO} \quad (\text{if } qe_{(c, cty)}^{FAO} > qm_{(c, cty)}^{FAO}) \quad (5.29)$$

where qm^{FAO} denotes the import quantity data from FAO, and qe^{FAO} denotes the export quantity data from FAO.

In the FBS dataset, global trade is not balanced; that is, total world exports and imports do not match. The discrepancy is retained as the signed accounting residual $qRes$, rather than interpreted as a physical transport loss.

$$qRes_c = \sum_{cty} qe_{(c, cty)} - \sum_{cty} qm_{(c, cty)} \quad (5.30)$$

5.2.5 Estimation of land constraints

Next, the estimation method for land areas unsuitable for agricultural use due to soil and topographical constraints is described. From the GAEZ dataset, area data for 57 agro-ecological zone (AEZ) classes, classified according to climate, soil, and topographical conditions, were obtained. Among these, land classes with strong constraints were selected and considered unsuitable for agricultural use. The area of such constrained land was calculated as follows:

$$qlCroCap_{(bas,cty)} = qlTot_{(bas,cty)}^{LUH2} \cdot \frac{\sum_{aezlim} qlGaez_{(aezlim,bas,cty)}}{\sum_{aez} qlGaez_{(aez,bas,cty)}} \quad (5.31)$$

where:

- $qlTot^{LUH2}$: total land area from LUH2,
- $qlGaez$: area data by AEZ class from GAEZ,
- aez : the set of 57 AEZ classes,
- $aezlim \in aez$: AEZ classes with severe land-use constraints.

In addition, protected-area information was incorporated using the World Database on Protected Areas (WDPA) [25]. The WDPA polygons were overlaid with basin-country units and classified into primary forest, secondary forest, other natural land, pasture, and rangeland. These areas form $\overline{qlProt}_{(bas,cty,lu_pa)}$, which is imposed as a lower bound on the corresponding land class; pasture and rangeland share the pasture constraint.

Urban land is based on the LUH2-consistent base-year area in the default model configuration. The base-year value is held constant in future years unless the optional SEDAC urban-projection extension is enabled. For this extension, the SEDAC 1-km urban fraction data [23, 24] were aggregated to basin-country units and converted to annual SSP-specific series. When the extension is switched on, these SEDAC series are used to update urban land in scenario runs while preserving the LUH2-consistent base-year level. To maintain physical consistency of the land balance, urban area was bounded by zero and by available land after subtracting wat/ice.

5.2.6 Estimation of forest carbon data

Forest and other natural-land carbon stocks are derived from spatial outputs of AIM-PLUM and VISIT and aggregated to basin-country units [5, 22]. Forest growth by stand age is used to calculate carbon sequestration from annual afforestation [28].

5.2.7 Adjustment of price data

From the FAO dataset, producer prices, import prices, and export prices for each country and commodity were obtained. Since the price data contained many outliers and missing values, appropriate preprocessing was conducted.

First, outlier values were treated as missing when the producer price was greater than ten times or less than one-tenth of the country-level median.

$$pp_{(c,cty)} \leftarrow \text{NA} \quad \text{if } (pp_{(c,cty)} > 10 \cdot \text{median}_{cty}(pp_{(c,cty)}) \cup pp_{(c,cty)} < 0.1 \cdot \text{median}_{cty}(pp_{(c,cty)})) \quad (5.32)$$

Next, missing values were complemented. A world average price was calculated from the available country-level producer prices, weighted by production quantities. A country-specific correction coefficient was then derived based on the deviation of each country's prices from the world average. This coefficient was used to impute the missing prices. The same procedure was applied for import and export prices.

$$pAvg_c = \frac{\sum_{cty} pp_{(c,cty)} \cdot qs_{(c,cty)}}{\sum_{cty} qs_{(c,cty)}} \quad (5.33)$$

$$pRelAvg_{cty} = \frac{\sum_c pp_{(c,cty)}}{pAvg_c} \quad (5.34)$$

$$pp_{(c,cty)} = pAvg_c \cdot pRelAvg_{cty} \quad (5.35)$$

where $pAvg$ denotes the global average producer price, and $pRelAvg$ denotes the country-specific adjustment coefficient.

Next, using the GTAP10 dataset, exogenous variables related to prices were estimated. The Producer Support Estimate (PSE) represents the monetary transfers from consumers and taxpayers to producers resulting from agricultural policies. In this study, the PSE is expressed as the proportionate impact of subsidies and taxes on producer prices. Similarly, the Consumer Support Estimate (CSE) represents the transfers from or to consumers, expressed as the proportionate impact of subsidies and taxes on consumer prices.

$$pse_{(i,cty)} = \frac{OSEP_{(i,cty)}^{GTAP} + \sum_i ISEP_{(i,ii,cty)}^{GTAP} - \sum_{endw} FTRV_{(i,endw,cty)}^{GTAP} + \sum_{endw} FBEP_{(i,endw,cty)}^{GTAP}}{VOA_{(i,cty)}^{GTAP}} \quad (5.36)$$

$$cse_{(i,cty)} = \frac{VDPM_{(i,cty)}^{GTAP} - VDP A_{(i,cty)}^{GTAP}}{VDPM_{(i,cty)}^{GTAP}} \quad (5.37)$$

where:

- $OSEP^{GTAP}$: subsidies/taxes on production,
- $ISEP^{GTAP}$: subsidies/taxes on intermediate inputs,
- $FTRV^{GTAP}$: taxes on factor inputs,
- $FBEF^{GTAP}$: subsidies on factor inputs,
- VOA^{GTAP} : production value,
- $VDPM^{GTAP}, VDPF^{GTAP}$: household consumption at market and consumer prices, respectively.

Indices: i : production sector; ii : intermediate input sector; $endw$: primary factor of production; cty : reporting country; $ccty$: partner country;

To link producer (farm-gate) prices with consumer (market) prices, the market margin ratio mmj was estimated. Intermediate inputs and factor inputs corresponding to market margins were identified from the GTAP input–output tables, and their total was divided by the production value to obtain the ratio. Similarly, trade-related costs mmm and mme were estimated from GTAP data, using the difference between import values at CIF prices and export values at FOB prices.

$$mmj_{(i,cty)} = \frac{\sum_{immj} VDF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} VIF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} EVF A_{(i,immj,cty)}^{GTAP}}{\left(\sum_{ii} VDF A_{(i,ii,cty)}^{GTAP} + \sum_{ii} VIF A_{(i,ii,cty)}^{GTAP} + \sum_{endw} EVF A_{(i,endw,cty)}^{GTAP} \right) - \left(\sum_{immj} VDF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} VIF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} EVF A_{(i,immj,cty)}^{GTAP} \right)}$$
(5.38)

$$mmm_{(i,ccty)} = \frac{\sum_{ccty} VIWS_{(i,ccty,ccty)}^{GTAP} - \sum_{ccty} VXWD_{(i,ccty,ccty)}^{GTAP}}{\sum_{ccty} VXWD_{(i,ccty,ccty)}^{GTAP}}$$
(5.39)

$$mme_{(i,cty)} = \frac{\sum_{ccty} VIWS_{(i,ccty,ccty)}^{GTAP} - \sum_{ccty} VXWD_{(i,ccty,ccty)}^{GTAP}}{\sum_{ccty} VXWD_{(i,ccty,ccty)}^{GTAP}}$$
(5.40)

where:

- $VDF A^{GTAP}, VIF A^{GTAP}, EVF A^{GTAP}$: intermediate and factor input values at consumer prices,
- $VIWS^{GTAP}, VXWD^{GTAP}$: import (CIF) and export (FOB) values,
- $immj \in (i \cup endw)$: goods and factors corresponding to market margins.

The import tariff (tm) and export tariff (te) were then calculated. The import tariff rate was obtained as the ratio of import value at domestic prices to that at international prices, and the export tariff rate as the ratio of export value at international prices to that at domestic prices.

$$tm_{(i,ccty)} = \frac{\sum_{ccty} VIMS_{(i,ccty,ccty)}^{GTAP}}{\sum_{ccty} VIWS_{(i,ccty,ccty)}^{GTAP}} \quad (5.41)$$

$$te_{(i,cty)} = \frac{\sum_{ccty} VXWD_{(i,cty,ccty)}^{GTAP}}{\sum_{ccty} VXMD_{(i,cty,ccty)}^{GTAP}} \quad (5.42)$$

where:

- $VIMS^{GTAP}$: import value at domestic (market) prices,
- $VXMD^{GTAP}$: export value at domestic (market) prices.

Finally, coefficients for decomposing producer prices by input factor were estimated. Each agricultural commodity is produced using land, feed, raw materials, and other production factors. Using the GTAP input–output tables, the cost share of each factor in the producer price was estimated. By multiplying the producer price by these shares, the base-year prices of each factor were obtained.

$$landCostShr_{(i,cty)} = \frac{\sum_{ilnd} VDF A_{(i,ilnd,cty)}^{GTAP} + \sum_{ilnd} VIF A_{(i,ilnd,cty)}^{GTAP} + \sum_{ilnd} EVF A_{(i,ilnd,cty)}^{GTAP}}{\left(\sum_{ii} VDF A_{(i,ii,cty)}^{GTAP} + \sum_{ii} VIF A_{(i,ii,cty)}^{GTAP} + \sum_{endw} EVF A_{(i,endw,cty)}^{GTAP} \right) - \left(\sum_{immj} VDF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} VIF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} EVF A_{(i,immj,cty)}^{GTAP} \right)} \quad (5.43)$$

$$feedCostShr_{(i,cty)} = \frac{\sum_{ifed} VDF A_{(i,ifed,cty)}^{GTAP} + \sum_{ifed} VIF A_{(i,ifed,cty)}^{GTAP} + \sum_{ifed} EVF A_{(i,ifed,cty)}^{GTAP}}{\left(\sum_{ii} VDF A_{(i,ii,cty)}^{GTAP} + \sum_{ii} VIF A_{(i,ii,cty)}^{GTAP} + \sum_{endw} EVF A_{(i,endw,cty)}^{GTAP} \right) - \left(\sum_{immj} VDF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} VIF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} EVF A_{(i,immj,cty)}^{GTAP} \right)} \quad (5.44)$$

$$intCostShr_{(i,cty)} = \frac{\sum_{iimd} VDF A_{(i,iimd,cty)}^{GTAP} + \sum_{iimd} VIF A_{(i,iimd,cty)}^{GTAP} + \sum_{iimd} EVF A_{(i,iimd,cty)}^{GTAP}}{\left(\sum_{ii} VDF A_{(i,ii,cty)}^{GTAP} + \sum_{ii} VIF A_{(i,ii,cty)}^{GTAP} + \sum_{endw} EVF A_{(i,endw,cty)}^{GTAP} \right) - \left(\sum_{immj} VDF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} VIF A_{(i,immj,cty)}^{GTAP} + \sum_{immj} EVF A_{(i,immj,cty)}^{GTAP} \right)} \quad (5.45)$$

$$1 - ipt2_{(i,cty)} = 1 - (landCostShr_{(i,cty)} + feedCostShr_{(i,cty)} + intCostShr_{(i,cty)}) \quad (5.46)$$

where:

- $landCostShr$: share of land cost in the producer price,

- $feedCostShr$: share of feed cost in the producer price,
- $intCostShr$: share of raw material cost in the producer price,
- $1 - ipt2$: share of other production factors in the producer price,
- $ilnd, ifed, iimd \in (i \cup endw)$: goods and factors corresponding to land, feed, and raw materials,
- The denominator represents the total value of domestic and imported intermediate inputs and factor inputs, excluding those corresponding to market margin sectors.

6 Parameter Estimation

6.1 Parameter estimation through regression analysis

This section presents the results of regression analyses used to estimate the parameters of the model. In the following equations, variables with an overline represent the dependent and independent variables used in the regressions.

Since price elasticities in the model were assumed to vary with per-capita GDP, the following regression equation was estimated:

$$\overline{elaP}_{(c,cty)} = \alpha_c^{pe} + \beta_c^{pe} \cdot \log(\overline{gdpPc}_{cty}) \quad (6.1)$$

For livestock products, it was assumed that production systems change with economic development. Thus, the parameters determining the production share were estimated using logistic regression analysis. In this estimation, dummy variables were introduced to represent climatic zones.

$$\log \left(\frac{\overline{sysShr}_{(a_{liv},bas,cty,cz,sys)}}{1 - \overline{sysShr}_{(a_{liv},bas,cty,cz,sys)}} \right) = \delta_{(a_{liv},cz,sys)}^{sys} + \rho_{(a_{liv},sys)}^{sys} \cdot \log(\overline{gdpPc}_{cty}) \quad (6.2)$$

It was further assumed that feed requirements and the share of concentrate feed also change with the level of economic development. Regression analyses were conducted using the following equations:

$$\log \left(\overline{feedReq}_{(a_{liv},bas,cty,cz,sys)} \right) = \alpha_{(a_{liv},cz,sys)}^{fe} + \beta_{(a_{liv},cz,sys)}^{fe} \cdot \log(\overline{gdpPc}_{cty}) \quad (6.3)$$

$$\log \left(\frac{\overline{feedConcShr}_{(a_{liv},bas,cty,cz,sys)}}{1 - \overline{feedConcShr}_{(a_{liv},bas,cty,cz,sys)}} \right) = \delta_{(a_{liv},cz,sys)}^{fs} + \rho_{(a_{liv},sys)}^{fs} \cdot \log(\overline{gdpPc}_{cty}) \quad (6.4)$$

Following Harmsen et al. (2019), the elasticity parameter σ related to the adoption of emission-reduction technologies was estimated using the following regression equation:

$$\log \left(1 - \frac{\overline{RRed}_{(src,gas,cty)}}{\overline{mrP}_{(src,gas,cty)}} \right) = \sigma_{(src,gas,cty)} \cdot \log(\overline{pCar}_{cty} + 1) \quad (6.5)$$

6.2 Results of parameter estimation

This section presents the results of the regression analyses conducted to estimate the model parameters. Variables with an underline in the equations represent the data used in the regressions.

6.2.1 Relationship between price elasticity and per-capita GDP

Figure 3 illustrates the relationship between per-capita GDP and price elasticity. Table 7 shows the estimated parameters and the coefficient of determination (R^2).

Since the model assumes that price elasticity varies with per-capita GDP, the parameters α_c^{pe} and β_c^{pe} were estimated using the regression model given in Eq. (6.1).

For all commodities, the estimated elasticities increased with higher levels of per-capita GDP. For most commodities, high coefficients of determination ($R^2 \approx 0.8$) were obtained. However, for commodities such as sugars, eggs, and spices, R^2 values were lower, around 0.5 or 0.3. The p -values for all regressions were below 0.001, indicating statistically significant relationships between the variables. Furthermore, the 95% confidence intervals for β^{pe} did not include zero, confirming the statistical significance of the estimated coefficients.

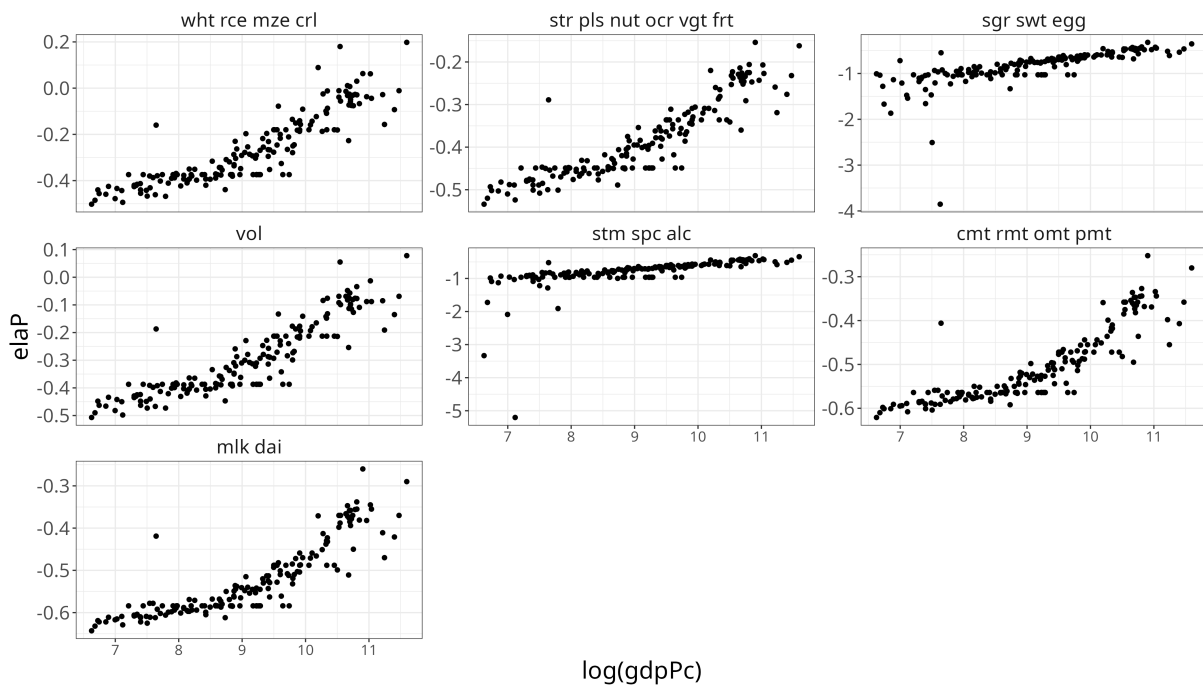


Figure 3: Relationship between per-capita GDP and price elasticity

Table 7: Estimated parameters of price elasticity from regression analysis

Commodity	α_{pe}	β_{pe}	β_{pe} (95% CI)		R^2	p -value
			lower	upper		
wht rce mze crl	−1.28	0.11	0.10	0.12	0.79	< 0.001
str pls nut ocr vgt frt	−0.99	0.07	0.06	0.07	0.81	< 0.001
sgr swt egg	−2.83	0.22	0.18	0.25	0.48	< 0.001
vol	−1.18	0.10	0.09	0.10	0.80	< 0.001
stm spc alc	−2.79	0.22	0.17	0.27	0.33	< 0.001
cmt rmt omt pmt	−1.04	0.06	0.05	0.06	0.76	< 0.001
mlk dai	−1.08	0.06	0.06	0.07	0.76	< 0.001

6.2.2 Relationship between livestock production system share and per-capita GDP

Figure 4 illustrates the relationship between per-capita GDP and the proportion of pasture-based production by livestock type and climatic zone. Table 8 presents the estimated parameters.

The parameters $\delta_{(a_{liv,cz,sys})}^{sys}$ and $\rho_{(a_{liv,sys})}^{sys}$ were estimated using the logistic regression model in Eq. (6.2).

As per-capita GDP increases, the share of pasture-based production decreases for all livestock products. Adjusted coefficients of determination were moderate (0.2–0.3), and all p -values were below 0.001, indicating statistically significant relationships. The 95% confidence intervals for ρ^{sys} did not include zero, confirming the significance of the coefficients.

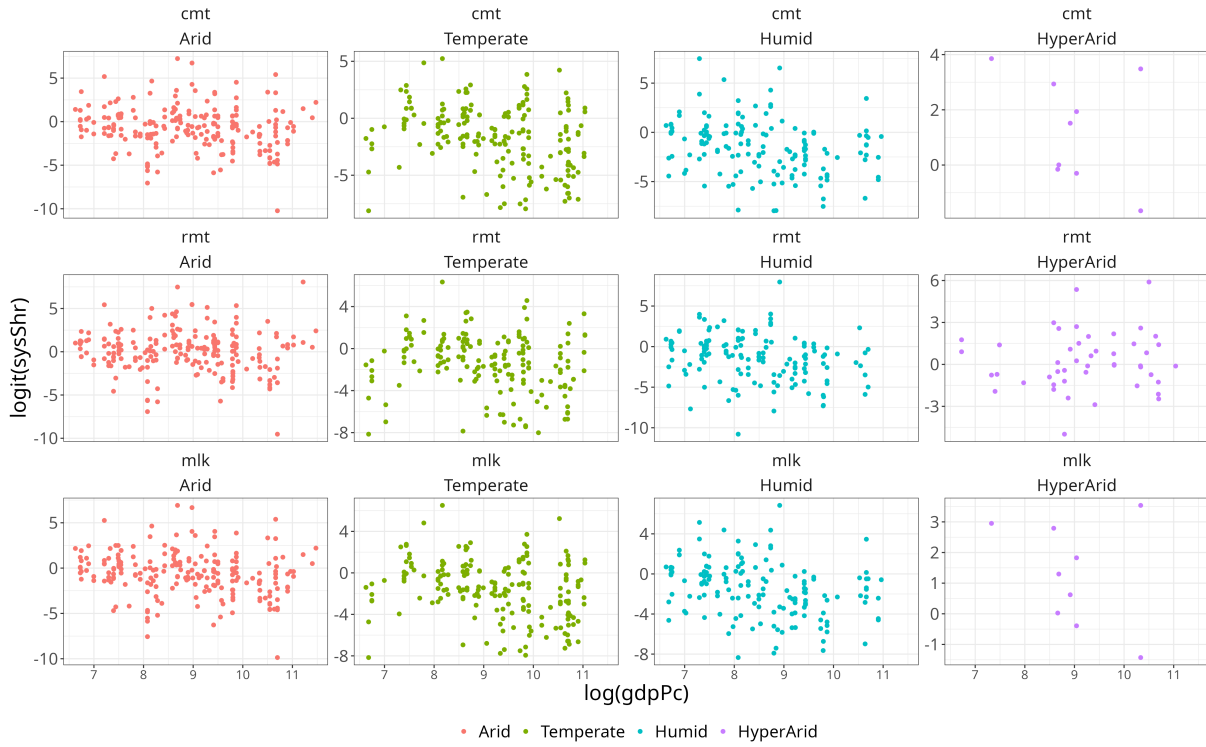


Figure 4: Relationship between per-capita GDP and share of pasture-based production

Table 8: Estimated parameters for pasture-based production share (95% confidence intervals in parentheses)

Commodity	ρ_{sys}	δ_{sys} by climate zone				R^2	p -value
		Arid	Temperate	Humid	HyperArid		
cmt	-0.51 (-0.69--0.33)	3.87	3.07	2.86	5.88	0.24	< 0.001
rmt	-0.34 (-0.53--0.15)	3.05	1.66	1.46	3.38	0.15	< 0.001
mlk	-0.54 (-0.71--0.36)	4.15	3.36	3.03	6.06	0.24	< 0.001

6.2.3 Relationship between feed requirements and per-capita GDP

Figure 5 shows the relationship between per-capita GDP and feed requirements by livestock type, climate zone, and production system. Table 9 presents the estimated regression parameters, obtained using the model in Eq. (6.3).

Higher per-capita GDP is associated with lower feed requirements. For ruminants, the adjusted R^2 ranged from 0.66 to 0.97, indicating high explanatory power, while for poultry and other livestock it was relatively low (0.22 and 0.05, respectively). The coefficients β^{fe} were smaller for ruminants, suggesting stronger efficiency improvements with economic development. All p -values were below 0.001, and the 95% confidence intervals for α^{fe} did not include zero, confirming statistical significance.

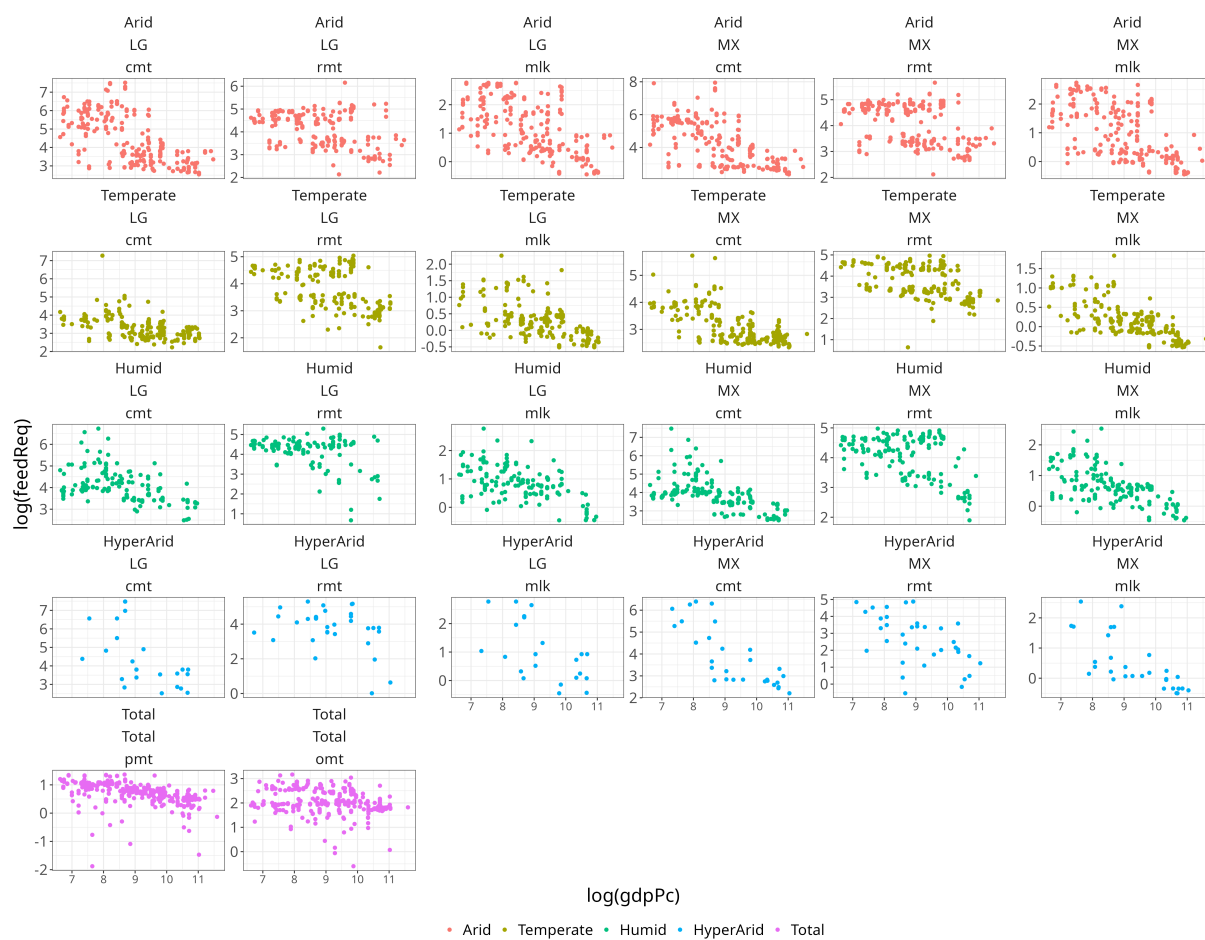


Figure 5: Relationship between per-capita GDP and feed requirements

Table 9: Estimated parameters of feed requirements (95% confidence intervals in parentheses)

Commodity	System	β_{fe}	α_{fe} by climate zone					R^2	p -value
			Arid	Temperate	Humid	HyperArid	Total		
cmt	LG	$-0.47 (-0.52--0.41)$	8.58	7.65	8.07	8.80	0.96	< 0.001	
rmt	LG	$-0.27 (-0.32--0.21)$	6.47	6.23	6.40	6.21	0.97	< 0.001	
mlk	LG	$-0.37 (-0.41--0.33)$	4.45	3.69	4.01	4.49	0.71	< 0.001	
cmt	MX	$-0.52 (-0.57--0.48)$	8.86	7.97	8.44	8.70	0.96	< 0.001	
rmt	MX	$-0.33 (-0.38--0.28)$	7.01	6.76	6.96	5.61	0.97	< 0.001	
mlk	MX	$-0.38 (-0.41--0.34)$	4.36	3.62	3.92	3.99	0.66	< 0.001	
pmt	Total	$-0.14 (-0.17--0.11)$					2.02	0.21 < 0.001	
omt	Total	$-0.09 (-0.13--0.04)$					2.82	0.05 < 0.001	

6.2.4 Relationship between concentrate feed share and per-capita GDP

Figure 6 shows the relationship between per-capita GDP and the share of concentrate feed by livestock type, climatic zone, and production system. Table 10 reports the estimated regression parameters, based on

the logistic specification in Eq. (6.4). Poultry and other livestock are excluded from the analysis because they do not consume pasture.

The results indicate that higher per-capita GDP is associated with a higher share of concentrate feed and a lower share of pasture. Adjusted coefficients of determination range from 0.74 to 0.90, indicating a relatively high explanatory power. All p -values are below 0.001, suggesting statistically significant relationships between the variables. Furthermore, the 95% confidence intervals for ρ^{fs} do not include zero, which confirms the statistical significance of the estimated coefficients.

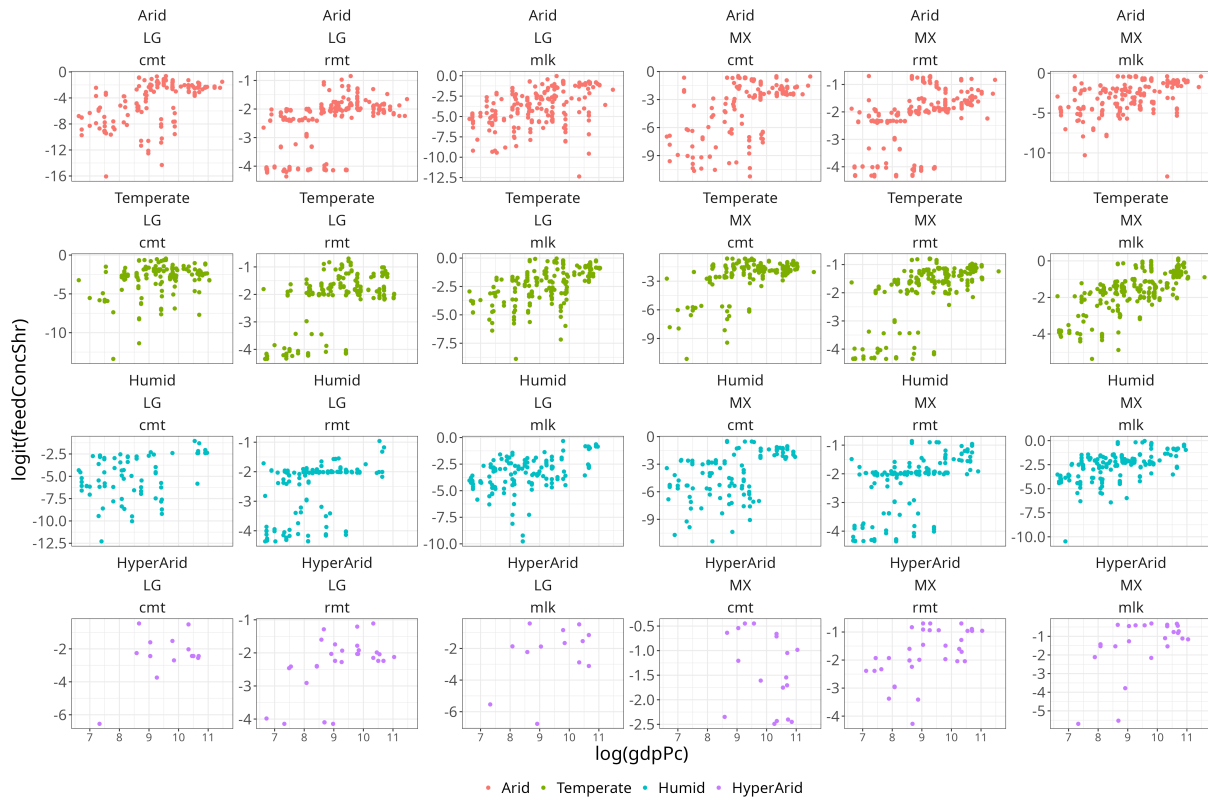


Figure 6: Relationship between per-capita GDP and concentrate feed share

Table 10: Estimated parameters of concentrate feed share (95% confidence intervals in parentheses)

Commodity	System	ρ_{fs}	δ_{fs} by climate zone				R^2	p -value
			Arid	Temperate	Humid	HyperArid		
cmt	LG	0.95 (0.74–1.16)	–13.24	–11.94	–13.09	–11.49	0.76	< 0.001
rmt	LG	0.42 (0.36–0.48)	–6.25	–6.00	–6.11	–6.36	0.90	< 0.001
mlk	LG	0.77 (0.65–0.89)	–10.68	–9.44	–9.99	–9.62	0.79	< 0.001
cmt	MX	1.04 (0.88–1.21)	–13.51	–12.21	–13.29	–11.89	0.74	< 0.001
rmt	MX	0.52 (0.46–0.57)	–6.89	–6.69	–6.81	–6.56	0.88	< 0.001
mlk	MX	0.70 (0.61–0.78)	–9.11	–8.05	–8.59	–8.06	0.78	< 0.001

7 Model Calculation Flow and Execution

7.1 Model Calculation Flow

Figure 7 shows the processing flow of the AIM-ALPHA model. The model begins with the processing of the original data sources, including extraction of necessary information, aggregation of gridded datasets, and conversion of data formats. Next, the processed data are adjusted through parameter estimation via regression analysis, data calibration to ensure equilibrium in the base year, and coefficient calibration for model consistency.

Using these prepared data, the model then performs future scenario calculations. For each year, equilibrium solutions are iteratively computed based on the base-year data, the results of the previous year, and the scenario assumptions. The aggregated annual results provide the final outputs for each scenario. Additionally, model results are converted into the submission templates of AgMIP (Agricultural Model Intercomparison and Improvement Project) and IAMC (Integrated Assessment Modeling Consortium) to facilitate data exchange and comparison. When enabled in `model_setting.sh`, the `AgMIPAssessment` submodule is also executed to run vetting and variable-consistency checks, and to generate plots and summary tables.

Optional pre-run bridges convert outputs from AIM-Hub, AFOLUB, and AIM-PLUM into AIM-ALPHA inputs. The switches `use_hub_output`, `use_afolu_output`, and `use_plum_output` control these bridges. When a scenario suffix is specified by `ModelInt`, the corresponding file in `setting/modelint` defines the linked-model setting.

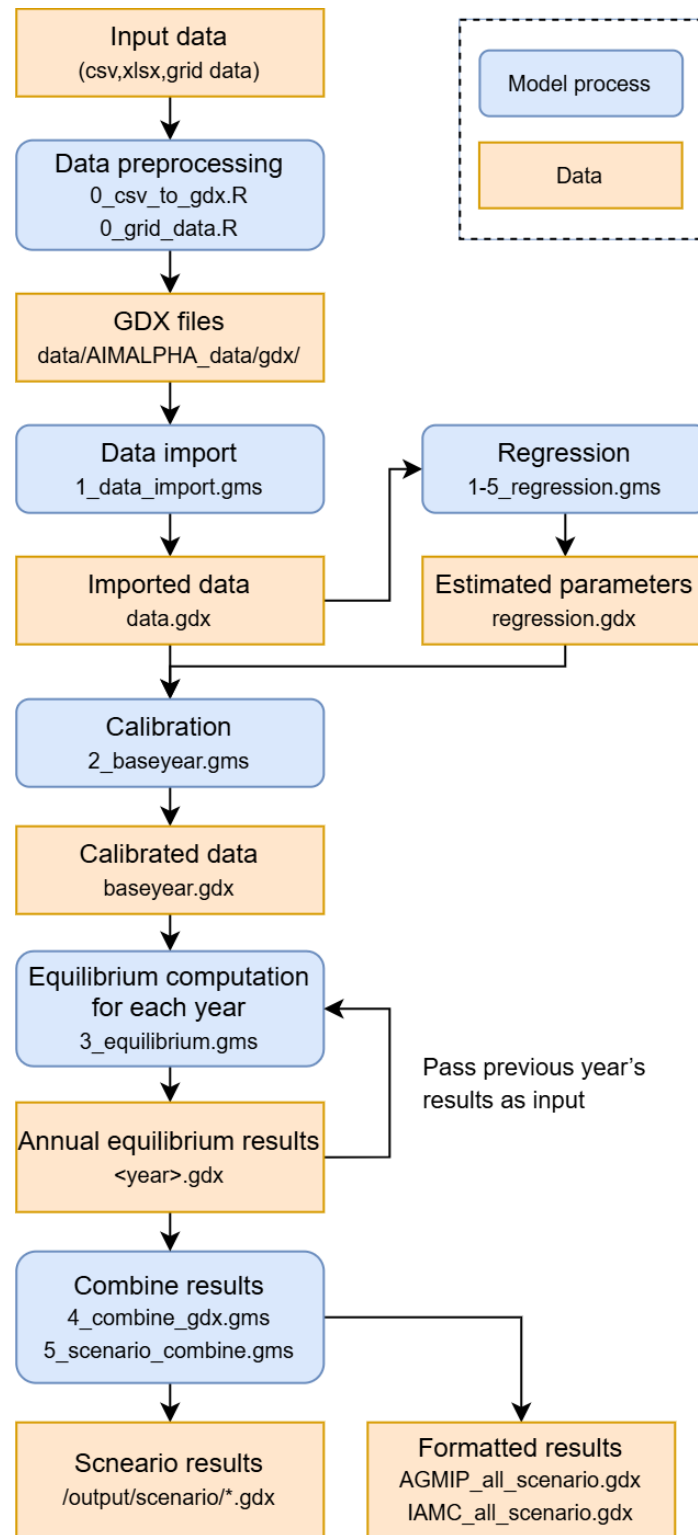


Figure 7: Processing flow of the AIM-ALPHA model

7.2 Programs within the Model

This section provides an overview of the major programs used in the AIM-ALPHA model. Each component plays a specific role in data processing, parameter calibration, and scenario simulation.

`0_csv_to_gdx.R`

Converts datasets in `csv` or `xlsx` format into `gdx` format. This script processes raw datasets such as FAOSTAT and MAC Curve sources.

`0_grid_data.R`

Aggregates gridded datasets by country and production unit, and exports the results in `gdx` format. It processes geospatial data from sources such as GAEZ, Livestock System, and LUH2.

`1_data_import.gms`

Integrates and reformats the `gdx`-formatted datasets into structures compatible with the model's computation framework. It performs country code conversions, consolidates data into AIM-ALPHA's food and land-use classifications, and includes submodules that output historical data in AgMIP and IAMC template formats.

`1-5_regression.gms`

Performs parameter regression analyses based on the original datasets to estimate elasticities and adjustment coefficients.

`2_baseyear.gms`

Adjusts baseline datasets to ensure equilibrium consistency in the base year.

`3_equilibrium.gms`

Solves for annual equilibrium states for future years, using base-year data and previous results as inputs.

`4_combine_gdx.gms`

Aggregates annual simulation outputs and compiles them into scenario-level result files.

`5_scenario_combine.gms`

Combines outputs from all scenarios to produce the comprehensive simulation dataset.

`Vis_each_scenario.R`

Generates graphical outputs and visualization figures from the simulation results.

`tool/AgMIPAssessment/shell/execution.sh`

Executes the AgMIPAssessment submodule, using `AGMIP_all_scenario.gdx` and historical templates for vetting, variable-consistency checks, and plot/table generation.

7.3 Execution Method

The AIM-ALPHA model can be executed by running the shell script `shell/ALPHA_exe.sh` on either Cygwin or Ubuntu environments.

Model configurations are defined in `shell/settings/model_setting.sh`. As summarized in Table 11, the configurable items include the selection of executable programs, scenario settings, and computational environment settings.

Basic execution

To run the model, move to the `shell` directory and execute:

```
cd shell
bash ALPHA_exe.sh
```

When `ALPHA_exe.sh` is called without any arguments, the default configuration file `model_setting.sh` is loaded automatically and the model is executed using those settings.

Using a custom setting file

It is also possible to specify an alternative setting file as an argument. In this case, the configuration defined in the specified file is used instead of `model_setting.sh`:

```
cd shell
bash ALPHA_exe.sh mysetting
```

Here, `mysetting.sh` denotes a user-defined setting file (e.g. a copy of `model_setting.sh` with modified options). Each run of `bash ALPHA_exe.sh <setting_file>` loads the corresponding setting file and executes the model according to those configurations.

Table 11: Model configuration parameters in `model_setting.sh`

Setting item	Description	Default
Model switches		
<code>data_download</code>	Executes a script to download data from submodules when the repository is first cloned. Once downloaded, this can be turned off.	on
<code>data_import</code>	Executes <code>1_data_import.gms</code> and <code>1-5_regression.gms</code> .	on
<code>baseyear_run</code>	Executes <code>2_baseyear.gms</code> .	on
<code>model_run</code>	Executes <code>3_equilibrium.gms</code> .	on
<code>combine_gdx</code>	Executes <code>4_combine_gdx.gms</code> .	on
<code>scenario_combine</code>	Executes <code>5_scenario_combine.gms</code> .	on
<code>Visualization</code>	Master switch for the plotting step; the <code>Vis_*</code> switches select what is drawn.	on
<code>AgMIP_assessment</code>	Executes AgMIPAssessment vetting and variable-consistency checks, and outputs plots and tables.	off
<code>csv2gdx_update</code>	Executes <code>0_csv_to_gdx.R</code> . If no data update is needed, this may be turned off.	off
<code>grid2gdx_update</code>	Executes <code>0_grid_data.R</code> . If no data update is needed, this may be turned off. If executed, the required gridded input files must be placed in <code>data</code> .	off
<code>use_hub_output</code>	Executes the Hub2ALPHA pre-run bridge.	off
<code>use_afolu_output</code>	Executes the AFOLU2ALPHA pre-run bridge.	off
<code>use_plum_output</code>	Executes the PLUM2ALPHA pre-run bridge.	off
<code>backcasting_run</code>	Runs the model backward from the base year to 2000.	off
<code>history_diagnose</code>	Compares the conditioned historical period with model results.	off
<code>surplus_run</code>	Calculates surplus changes relative to <code>surplus_reference</code> .	off
<code>report_year_interval</code>	Sets the reporting interval to one or five years.	5
<code>Vis_each_scenario</code>	Generates figures for individual scenarios.	on
<code>Vis_scenario_compare</code>	Enables comparison across scenarios in plots.	off
<code>Vis_scenario_list</code>	Restricts scenario-comparison plots to the listed scenarios; an empty list uses all scenarios.	empty
<code>Vis_model_compare</code>	Enables comparison across models in plots.	off
<code>Vis_history</code>	Compares simulated and historical data in plots.	off
<code>Vis_country, Vis_global17, Vis_map</code>	Selects country, regional, and map outputs.	on, on, off
<code>aggregate_global17</code>	Performs aggregation of results into 17-region, 10-region, and 5-region groupings.	on
<code>AgMIP_output</code>	Outputs results in AGMIP and IAMC_Template formats.	on
Scenario settings		
<code>socioeconomics</code>	Specifies the socioeconomic scenario. A file with the scenario name should exist in <code>setting/socioeconomic</code> .	SSP2
<code>climatepolicies</code>	Specifies the climate-policy scenario. A file with the scenario name should exist in <code>setting/climatepolicy</code> .	BaU
<code>climates</code>	Specifies the climate scenario. A file with the scenario name should exist in <code>setting/climate</code> .	NoCC
<code>runlist</code>	Allows direct scenario specification without using the scenario matrix. Scenario names are defined as <code>socioeconomic climatepolicy climate</code> (e.g., <code>SSP2 500C_CACN_DAC NoCC</code>). If this item is specified, the three settings above are ignored.	–
<code>start_year</code>	Starting year of simulation.	2015
<code>target_year</code>	Ending year of simulation.	2100
Execution environment		
<code>run_parallel</code>	Runs multiple scenarios in parallel.	on
<code>serial_iterations</code>	Ordered list of solver iteration settings tried when <code>run_parallel</code> is off.	0, 3, 5
<code>MultiSol</code>	Specifies the number of solver candidates run concurrently for each year when <code>run_parallel</code> is on; the first successful solution is retained.	3
<code>ModelInt</code>	Optional suffix appended to scenario IDs and linked to <code>setting/modelint/<ModelInt>.gms</code> . If empty, no suffix is appended.	""
<code>NCPU</code>	Maximum number of CPUs used for parallel computation.	8
<code>year_timeout</code>	Maximum computation time allowed per simulation year (in seconds). If this time limit is exceeded, the scenario run is aborted.	900
<code>solprint</code>	Writes the full solve report into the listing files of <code>3_equilibrium.gms</code> , for debugging. Turning it off keeps the listing files small.	on

7.4 Execution Environment Requirements

The model can be executed under either Cygwin or Ubuntu with the following dependencies installed:

- **Cygwin** (required packages: `zip`, `unzip`)
- **Ubuntu**
- **R** (version $\geq 4.3.2$; required libraries: `"tidyverse"`, `"gdxrrw"`, `"openxlsx"`, `"readxl"`, `"ggplot2"`, `"ggpmisc"`, `"RColorBrewer"`, `"purrr"`, `"furrr"`, `"scales"`, `"patchwork"`, `"sf"`, `"raster"`, `"ncdf4"`, `"exactextractr"`, `"rmapshaper"`, `"rnaturalearth"`, `"rnaturalearthdata"`, `"grDevices"`)
- **GAMS** (version ≥ 38 , with the PATH solver available)
- **GitHub**

8 Development Guidelines

This section summarizes the conventions used when extending or maintaining AIM-ALPHA. The purpose is to keep the GAMS programs readable and to preserve consistency across calibration, equilibrium calculation, and output processing.

8.1 Location of Model Components

Sets and mappings shared by several programs are defined under `define`. Model assumptions and constant values are placed in the relevant file under `prog/inc_prog/const`. Their application conditions remain in the program that uses them. Source-data corrections are kept separate from economic equations and accompanied by a short explanation of the underlying data issue.

The model is processed in a sequence. Data are imported and regionalized in `1_data_import.gms` and `1-5_regression.gms`; base-year values are calibrated in `2_baseyear.gms`; and the dynamic equilibrium is solved in `3_equilibrium.gms`. Annual results are then combined and aggregated by the programs beginning with `4_` and `5_`. A change to a symbol or equation should therefore be followed through each affected stage, including the shell and R programs that read the resulting GDX files.

Numerical initialization, bounds, fixing, and scaling are maintained in `3_level.gms`, `3_lower.gms`, `3_fix.gms`, and `3_scale.gms`. Economic assumptions should not be hidden in these numerical-support files. The global `base_year` setting is used for calibration-year references instead of a year written directly into the program.

8.2 Symbol Names

Names are kept short enough for equations to remain readable, while retaining the economic meaning of each symbol. GAMS is case-insensitive, so capitalization is a convention for readers rather than a means of distinguishing symbols.

Table 13: General naming conventions

Symbol type	Convention	Examples
Variable	Begins with an uppercase letter	QD, QSBas, PL
Parameter	Begins with a lowercase letter	qd0, qsBasPrev1, elaP
Equation	Begins with an uppercase letter and ends in Eq	QDEq, CroZeroProfitEq
Set	Lowercase	c, cty, a_liv

Quantity variables begin with Q. The QD family contains total domestic demand and its components, such as QDFood, QDFed, and QDInt. Domestic production uses QS; production by basin, crop technology, and livestock system uses QSBas, QSWat, and QSSys, respectively. Land quantities use QL, and land prices use PL. Other common stems are P for prices, S for shares, R for rates, Y for yields, C for unit costs, and N for stocks or counts.

Short qualifiers are added in CamelCase when a distinction is needed. Examples include Bas for basin, Cty for country, Cro for crops, Liv for livestock, Wat for crop production technology, Pas for pasture, and Emis for emissions. A one-letter qualifier is avoided when it may have more than one interpretation.

Complementarity multipliers begin with Mu and name the associated constraint. Thus, MuBasMin is paired with QSBasMinEq, and MuAfrCap is paired with AfrCapEq. Equation names describe their role using terms such as Bal, Logit, Min, Max, Parity, and ZeroProfit.

8.2.1 Sets and Mappings

Basic Set names are lowercase and normally contain no more than three characters. A subset is named as the parent Set followed by its classification, as in a_cro, a_liv_rum, and c_ntrd. A two-dimensional mapping is named <left>_<right>_map; for example, cty_reg_map(cty,reg) maps countries to regions. Aliases use the same stem followed by 2, as in a_cro2.

8.2.2 Time and Processing State

The same stem is retained as a value moves through calibration and the annual calculation. The suffixes Prev1 and Prev2 identify stored states one and two annual steps earlier. A bare Prev denotes the one-period-lagged working value for the current substep; when both stored states are needed, it is normally interpolated from Prev2 and Prev1. Ref is reserved for a non-temporal benchmark or reference value, and Drv denotes an exogenous driver before the endogenous response is applied.

The suffix _load is reserved for data read from an external file before scenario or year selection. Path denotes a future path constructed inside the model, and Raw denotes a value before a bound or adjustment is applied. Bounds use the functional suffixes Cap, Min, and Max.

8.2.3 External Data and Output

Names fixed by an external GDX schema are not changed at the interface. When necessary, the symbol is renamed on the model side of the \$load statement. Established abbreviations in imported databases, including the GTAP parameters pse, cse, mmj, mmm, mme, tm, and te, are retained.

Annual GDX outputs use <stem>_result. Combined time-series values use <stem>_load, single-year scratch values use <stem>Single, and aggregated values use <stem>Agg_load. A value derived from an endogenous variable retains the uppercase variable stem; an output derived from a Parameter retains the lowercase Parameter stem.

Descriptions are written as short noun phrases and include units where applicable. Standard forms include [1000 t], [1000 ha], [1000 head], [kt CO₂-eq/yr], [USD/t], and [USD/ha].

8.3 Time Steps and Scenario Measures

The annual calculation can divide the transition from `pre_year` to `year` into smaller intervals. The loop in `3_equilibrium.gms` evaluates `3_timestep.gms`, applies the scenario measures in `3_scenario.gms`, and then solves the equilibrium. The solution levels from one pass provide the starting point for the next pass, which allows the model to approach the target year gradually when a direct annual change would be numerically difficult.

When `iter_num=0`, the loop is evaluated once and the requested year is solved directly. When `iter_num=k>0`, the loop runs from `n=0` to `n=k`. The first solve represents `pre_year`, the last represents `year`, and the intervening solves divide the transition into `k` equal intervals. There are therefore `k+1` equilibrium solves. The scalar `tCur` gives the time represented by the current pass, and the interpolation weights are

$$wPrev = \frac{k-n}{k}, \quad wCur = \frac{n}{k}, \quad tCur = pre_year + (year - pre_year)wCur. \quad (8.1)$$

Parameters that vary exogenously over time must be assigned on every pass through `3_timestep.gms`. A parameter stored at the two endpoint years is normally calculated as

$$x = x(pre_year)wPrev + x(year)wCur, \quad (8.2)$$

while a trend may instead be evaluated directly at `tCur`. This applies to drivers such as population, GDP per capita, elasticities, yield changes, carbon prices, and mitigation parameters. Quantities carried forward from earlier endogenous solutions use `Prev1` and `Prev2` for the stored annual states. Their one-period-lagged value at the current substep is

$$xPrev = xPrev2 wPrev + xPrev1 wCur. \quad (8.3)$$

At the final substep, and whenever `iter_num=0`, `xPrev` therefore equals `xPrev1`. Intermediate solves are used for numerical continuation and are not written as separate annual results; `3_result.gms` is evaluated only after the loop has finished.

The order of calculation is important. `3_timestep.gms` first restores or constructs the baseline value for the current time point, after which `3_scenario.gms` applies the active measures. If a scenario changes a Parameter in place, that Parameter must be restored or rebuilt on the next pass before the measure is applied again. Otherwise the policy effect compounds across substeps. Conversely, an exogenous Parameter that is set only outside the loop remains stale while the model advances toward the target year.

Introducing a time-varying exogenous driver therefore requires four related changes. Its annual series is declared and loaded outside the loop; its working Parameter is assigned in `3_timestep.gms` using `wPrev` and `wCur`, or directly from `tCur`; any fallback and backcasting behavior is specified there; and its baseline value is restored there if a scenario later modifies it in place. The implementation is checked at both endpoints: `n=0` must reproduce the `pre_year` value, `n=iter_num` must reproduce the target-year value, and `iter_num=0` must assign the target-year value directly.

A new scenario measure is added as an include file under `prog/inc_prog/scenario`. Its `SCE_*` switch is set to `off` by default, dispatched from `3_scenario.gms`, and enabled only in the relevant setting file. Symbols are declared outside the scenario include because that include is evaluated repeatedly.

Conditional compilation blocks use matching labels, for example `$ifthen.name` and `$endif.name`. Conditions over large domains include a sparse operand so that inactive tuples are not evaluated unnecessarily.

8.4 Validation of Changes

All affected GAMS stages are compiled with the settings used by the intended run. Changes to imported or exported symbols are tested through data import, base-year calibration, equilibrium calculation, and output aggregation. Modified shell and R programs are also parsed with their native tools.

For a rename or another change that is not intended to affect model behavior, a pre-change GDX is retained. Old and new symbols are mapped explicitly and their records are compared; both the domain and the values should agree exactly. A representative scenario run is then checked for GAMS execution errors, model status, solver status, and diagnostic residuals. A process exit code of zero alone does not establish that all annual solves succeeded.

Generated outputs, temporary solver files, and local configuration files are not included in a contribution unless they are an intentional part of the change.

9 Update History

2025-11-02	 	R. Totake	v1.0.0
Initial release of AIM-ALPHA documentation.			
2026-2-11	 	R. Totake	v1.0.1
Added protected areas; split forests into primary and secondary categories; and divided pastureland into grazed and ungrazed categories to more accurately represent land use and grazing intensity.			
2026-2-21	 	R. Totake	v1.0.2
Added an optional SSP-based urban land projection extension using the SEDAC 1-km urban dataset, while keeping the LUH2-consistent base-year urban area as the default.			
2026-2-27	 	R. Totake	v1.0.3
Updated scenario and inter-model integration settings: introduced split scenario controls (<code>socioeconomic_setting</code> , <code>climatepolicy_setting</code> , <code>climate_setting</code>), standardized scenario suffix control with <code>ModelInt</code> , changed <code>runlist</code> format to <code>socioeconomic climatepolicy climate</code> , and added optional pre-run bridge processes (<code>hubauto/Hub2ALPHA</code> and <code>afoluauto/AFOLU2ALPHA</code>).			
2026-08-29	 	R. Totake	v1.0.4
Updated the land-use and afforestation formulation, protected-area constraints, crop-yield drivers, inter-model integration, and model settings.			
2026-09-02	 	R. Totake	v1.1.0
Standardized model symbol names and documentation, added development guidelines, and updated the regression scripts and figures.			

References

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- [1] World Bank. *World Bank Official Boundaries*. Dataset. 2025. URL: <https://datacatalog.worldbank.org/search/dataset/0038272/World-Bank-Official-Boundaries>.
 - [2] FAO Land and Water Division. *Major Hydrological Basins of the World*. 2011. URL: <https://data.apps.fao.org/catalog/iso/7707086d-af3c-41cc-8aa5-323d8609b2d1>.
 - [3] Fujimori, S. et al. “SSP3: AIM Implementation of Shared Socioeconomic Pathways”. In: *Global Environmental Change* 42 (Jan. 2017), pp. 268–283. DOI: [10.1016/j.gloenvcha.2016.06.009](https://doi.org/10.1016/j.gloenvcha.2016.06.009).
 - [4] Hasegawa, T., Fujimori, S., Tsuchiya, K., Ito, A., and Takahashi, K. “National Land-based Climate Mitigation Scenarios Dataset”. In: *Scientific Data* 12.1 (July 2025), p. 1077. DOI: [10.1038/s41597-025-05427-5](https://doi.org/10.1038/s41597-025-05427-5).
 - [5] Hasegawa, T., Fujimori, S., Ito, A., Takahashi, K., and Masui, T. “Global Land-Use Allocation Model Linked to an Integrated Assessment Model”. In: *Science of The Total Environment* 580 (Feb. 2017), pp. 787–796. DOI: [10.1016/j.scitotenv.2016.12.025](https://doi.org/10.1016/j.scitotenv.2016.12.025).
 - [6] Crespo Cuaresma, J. “Income Projections for Climate Change Research: A Framework Based on Human Capital Dynamics”. In: *Global Environmental Change* 42 (Jan. 2017), pp. 226–236. DOI: [10.1016/j.gloenvcha.2015.02.012](https://doi.org/10.1016/j.gloenvcha.2015.02.012).
 - [7] KC, S. et al. *Wittgenstein Center (WIC) Population and Human Capital Projections - 2023*. Feb. 2024. DOI: [10.5281/zenodo.10618931](https://doi.org/10.5281/zenodo.10618931).
 - [8] Calvin, K. et al. “GCAM v5.1: Representing the Linkages between Energy, Water, Land, Climate, and Economic Systems”. In: *Geoscientific Model Development* 12.2 (Feb. 2019), pp. 677–698. DOI: [10.5194/gmd-12-677-2019](https://doi.org/10.5194/gmd-12-677-2019).
 - [9] FAOSTAT. URL: <https://www.fao.org/faostat/en/%5C#home>.
 - [10] FAO and IIASA. *Global Agro-Ecological Zones Version 4 (GAEZ V4)*. 2025. URL: <http://www.fao.org/gaez/>.
 - [11] Aguiar, A., Chepeliev, M., Corong, E., McDougall, R., and van der Mensbrugghe, D. “The GTAP Data Base: Version 10”. In: *Journal of Global Economic Analysis* 4.1 (2019), pp. 1–27. DOI: [10.21642/JGEA.040101AF](https://doi.org/10.21642/JGEA.040101AF).
 - [12] (IFPRI), I. F. P. R. I. *Global Spatially-Disaggregated Crop Production Statistics Data for 2020, Version 2.0*. 2024. DOI: [10.7910/DVN/SWPENT](https://doi.org/10.7910/DVN/SWPENT).
 - [13] Robinson, S. et al. *The International Model for Policy Analysis of Agricultural Commodities and Trade (IMPACT): Model Description for Version 3*. SSRN Scholarly Paper. Rochester, NY, Nov. 2015. DOI: [10.2139/ssrn.2741234](https://doi.org/10.2139/ssrn.2741234).

- [14] European Commission, Joint Research Centre (JRC) and International Energy Agency (IEA). *EDGAR (Emissions Database for Global Atmospheric Research) Community GHG Database: IEA-EDGAR CO₂, EDGAR CH₄, EDGAR N₂O, and EDGAR F-GASES, Version 2024*. 2024. URL: https://edgar.jrc.ec.europa.eu/dataset_ghg2024.
- [15] Herrero, M. et al. *Livestock Production Systems*. 2018. DOI: [10.4225/08/5aa068b33fe06](https://doi.org/10.4225/08/5aa068b33fe06).
- [16] Muhammad, A., Seale James L., J., Meade, B., and Regmi, A. *International Evidence on Food Consumption Patterns: An Update Using 2005 International Comparison Program Data*. Technical Bulletin No. TB-1929. U.S. Department of Agriculture, Economic Research Service, Feb. 2013.
- [17] Robinson, T. P. et al. *Global Distribution of Ruminant Livestock Production Systems V5 (5 Minutes of Arc)*. Sept. 2018. DOI: [10.7910/DVN/WPDSZE](https://doi.org/10.7910/DVN/WPDSZE).
- [18] Harmsen, J. H. M. et al. “Long-Term Marginal Abatement Cost Curves of Non-CO₂ Greenhouse Gases”. In: *Environmental Science & Policy* 99 (Sept. 2019), pp. 136–149. DOI: [10.1016/j.envsci.2019.05.013](https://doi.org/10.1016/j.envsci.2019.05.013).
- [19] Hurtt, G. C. et al. “Harmonization of Global Land Use Change and Management for the Period 850–2100 (LUH2) for CMIP6”. In: *Geoscientific Model Development* 13.11 (Nov. 2020), pp. 5425–5464. DOI: [10.5194/gmd-13-5425-2020](https://doi.org/10.5194/gmd-13-5425-2020).
- [20] Hanasaki, N. et al. “An Integrated Model for the Assessment of Global Water Resources – Part 1: Model Description and Input Meteorological Forcing”. In: *Hydrology and Earth System Sciences* 12 (2008), pp. 1007–1025. DOI: [10.5194/hess-12-1007-2008](https://doi.org/10.5194/hess-12-1007-2008).
- [21] Hanasaki, N. et al. “An Integrated Model for the Assessment of Global Water Resources – Part 2: Applications and Assessments”. In: *Hydrology and Earth System Sciences* 12 (2008), pp. 1027–1037. DOI: [10.5194/hess-12-1027-2008](https://doi.org/10.5194/hess-12-1027-2008).
- [22] Ito, A. and Inatomi, M. “Water-Use Efficiency of the Terrestrial Biosphere: A Model Analysis Focusing on Interactions between the Global Carbon and Water Cycles”. In: *Journal of Hydrometeorology* 13.2 (2012), pp. 681–694. DOI: [10.1175/JHM-D-10-05034.1](https://doi.org/10.1175/JHM-D-10-05034.1).
- [23] Gao, J. and Pesaresi, M. *Global 1-km Downscaled Urban Land Extent Projection and Base Year Grids by SSP Scenarios, 2000-2100*. Palisades, NY, 2021. DOI: [10.7927/1z4r-ez63](https://doi.org/10.7927/1z4r-ez63). URL: <https://doi.org/10.7927/1z4r-ez63>.
- [24] Gao, J. and Pesaresi, M. “Downscaling SSP-consistent Global Spatial Urban Land Projections from 1/8-degree to 1-km Resolution 2000-2100”. In: *Scientific Data* 8.1 (2021), p. 253. DOI: [10.1038/s41597-021-01052-0](https://doi.org/10.1038/s41597-021-01052-0).
- [25] IUCN and UNEP-WCMC. *The World Database on Protected Areas (WDPA) [On-line]*, December 2025. Cambridge, UK, Dec. 2025. URL: <https://www.protectedplanet.net>.
- [26] FAO. *Global food losses and food waste*. 2011. URL: <https://www.fao.org/4/mb060e/mb060e00.htm>.
- [27] Wu, W. et al. “Global Advanced Bioenergy Potential under Environmental Protection Policies and Societal Transformation Measures”. In: *GCB Bioenergy* 11.9 (2019), pp. 1041–1055. DOI: [10.1111/gcbb.12614](https://doi.org/10.1111/gcbb.12614).

- [28] Hasegawa, T., Fujimori, S., Ito, A., and Takahashi, K. “Careful Selection of Forest Types in Afforestation Can Increase Carbon Sequestration by 25% without Compromising Sustainability”. In: *Communications Earth & Environment* 5.1 (Apr. 2024), pp. 1–10. doi: [10.1038/s43247-024-01336-4](https://doi.org/10.1038/s43247-024-01336-4).

Appendix A: Production Units

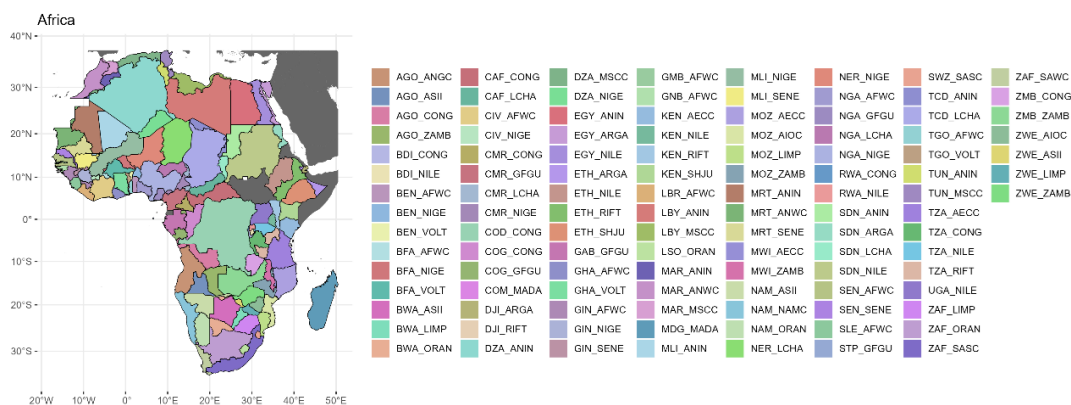


Figure A1: Production units in Africa

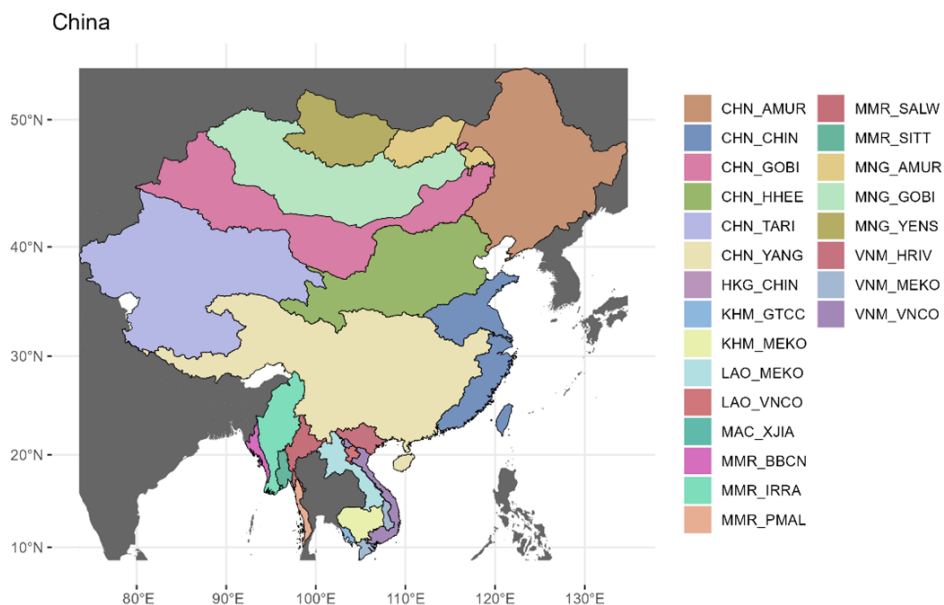


Figure A2: Production units in the China region

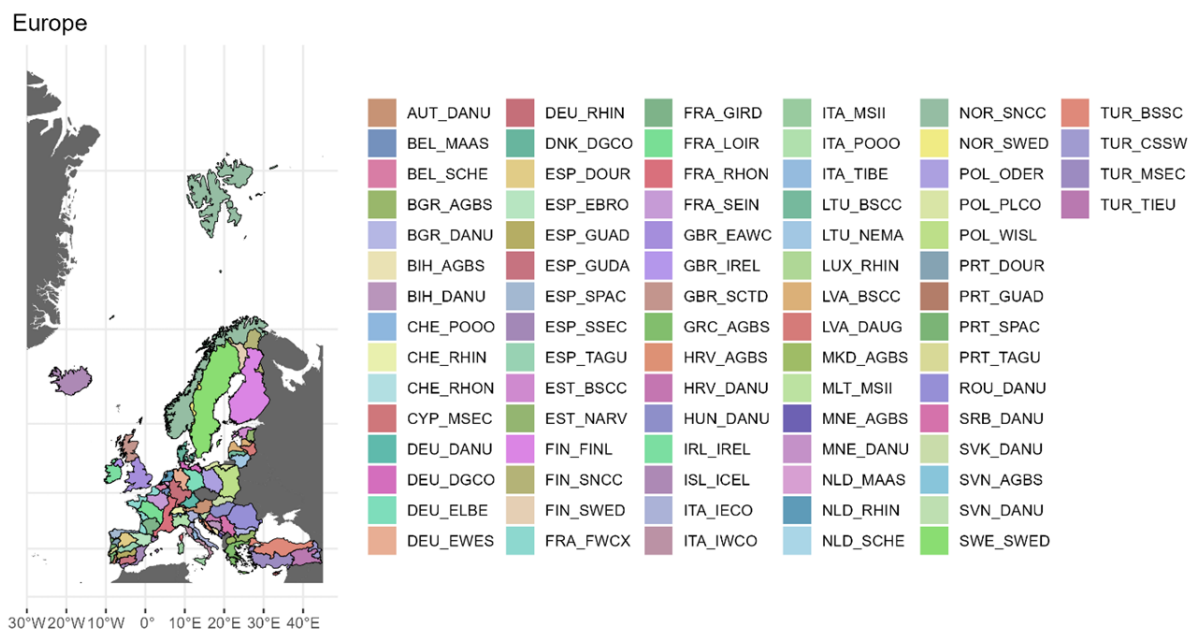


Figure A3: Production units in Europe

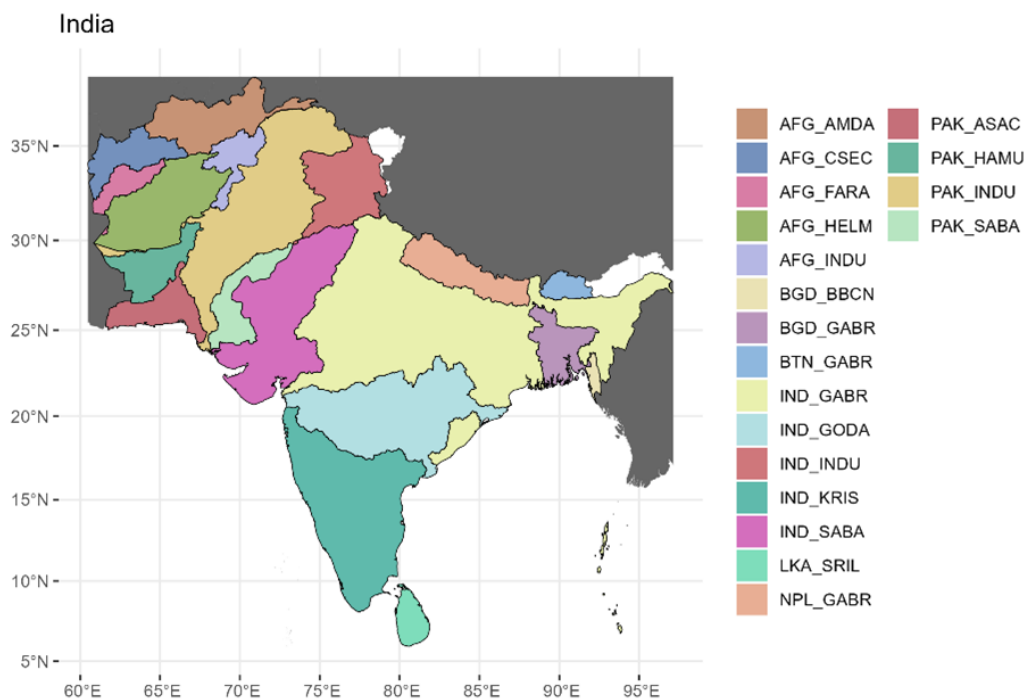


Figure A4: Production units in the India region

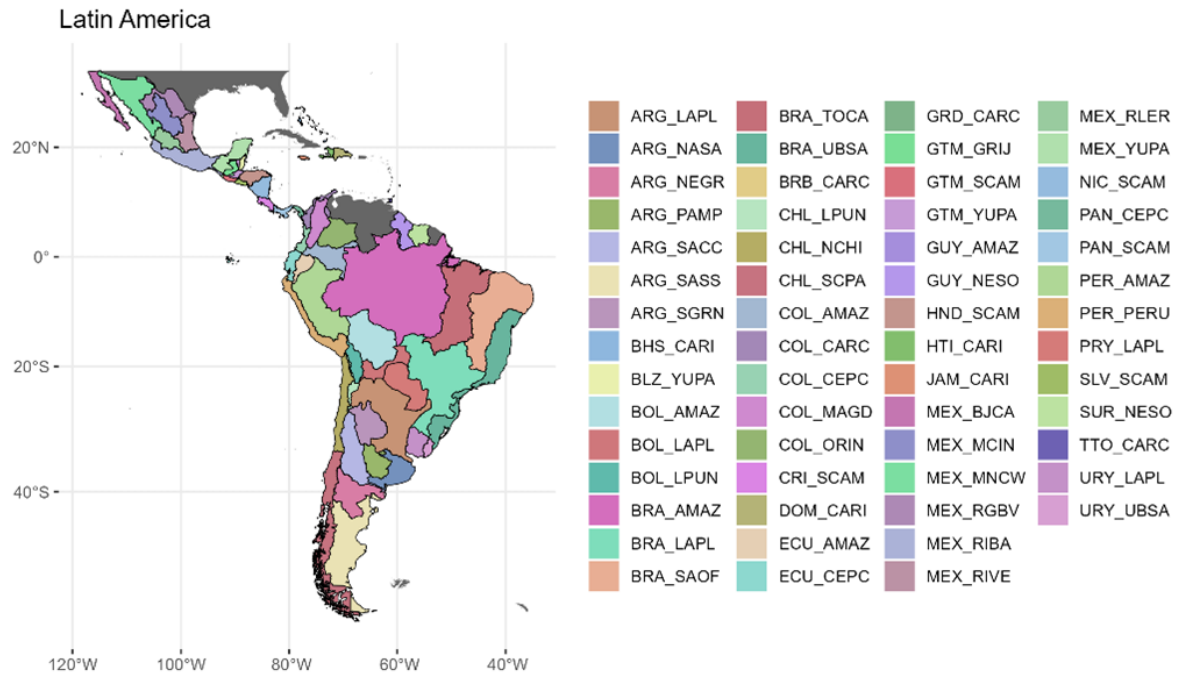


Figure A5: Production units in Latin America

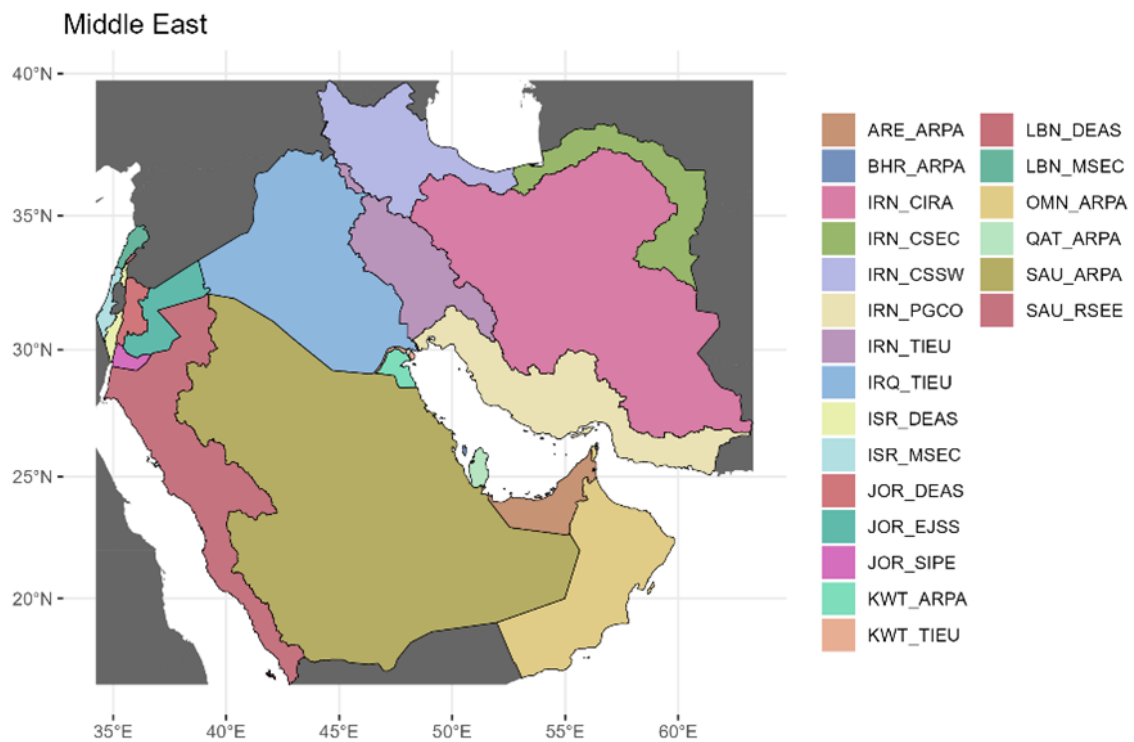


Figure A6: Production units in the Middle East

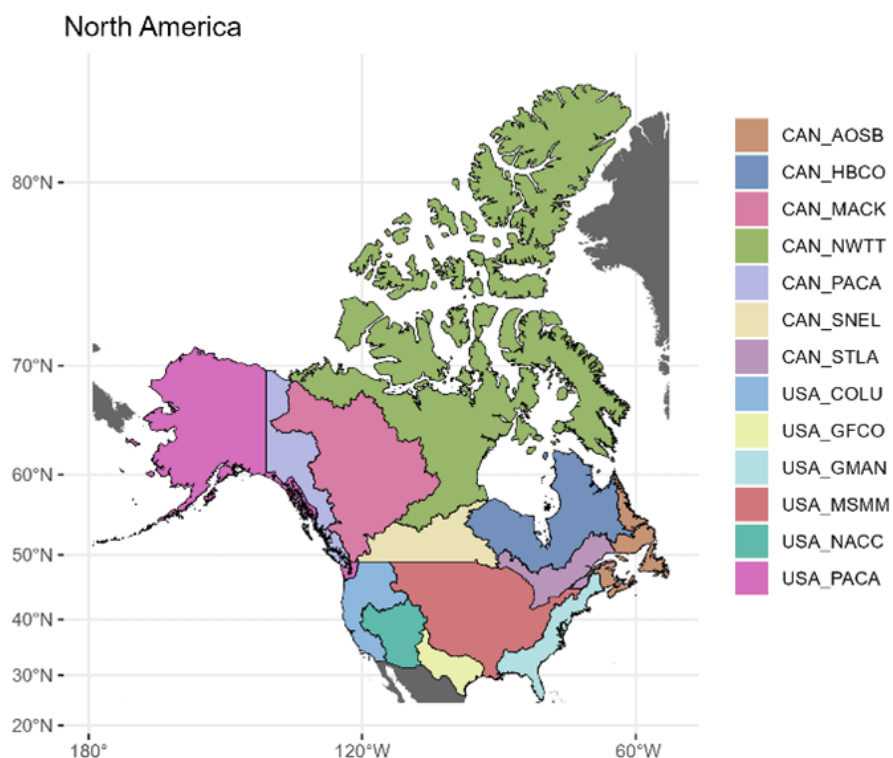


Figure A7: Production units in North America

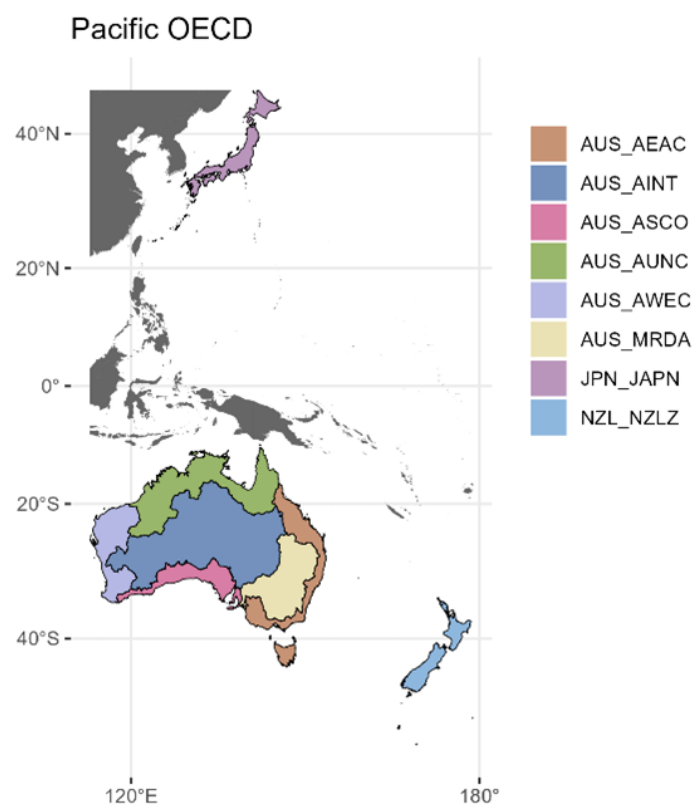


Figure A8: Production units in the Pacific OECD region

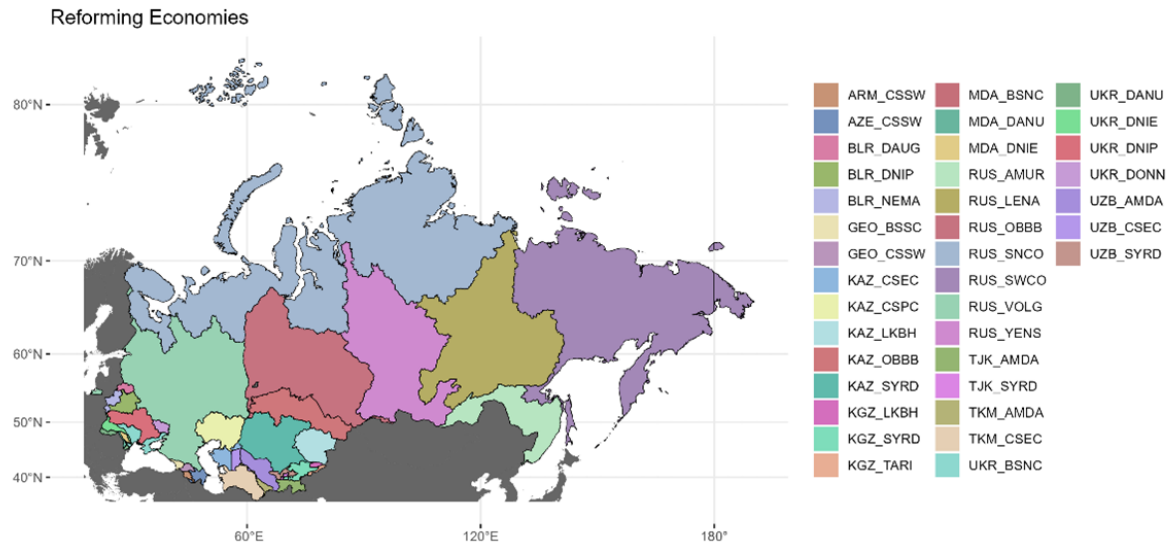


Figure A9: Production units in the former Soviet Union

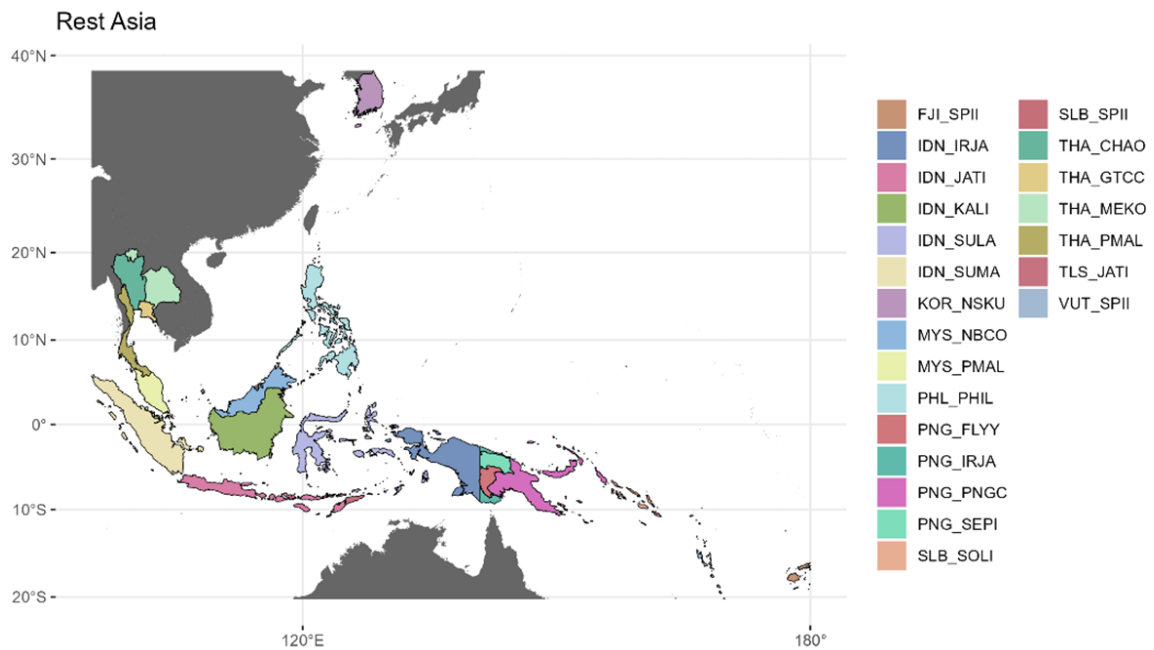


Figure A10: Production units in other Asian regions

Appendix B: Regional Aggregation

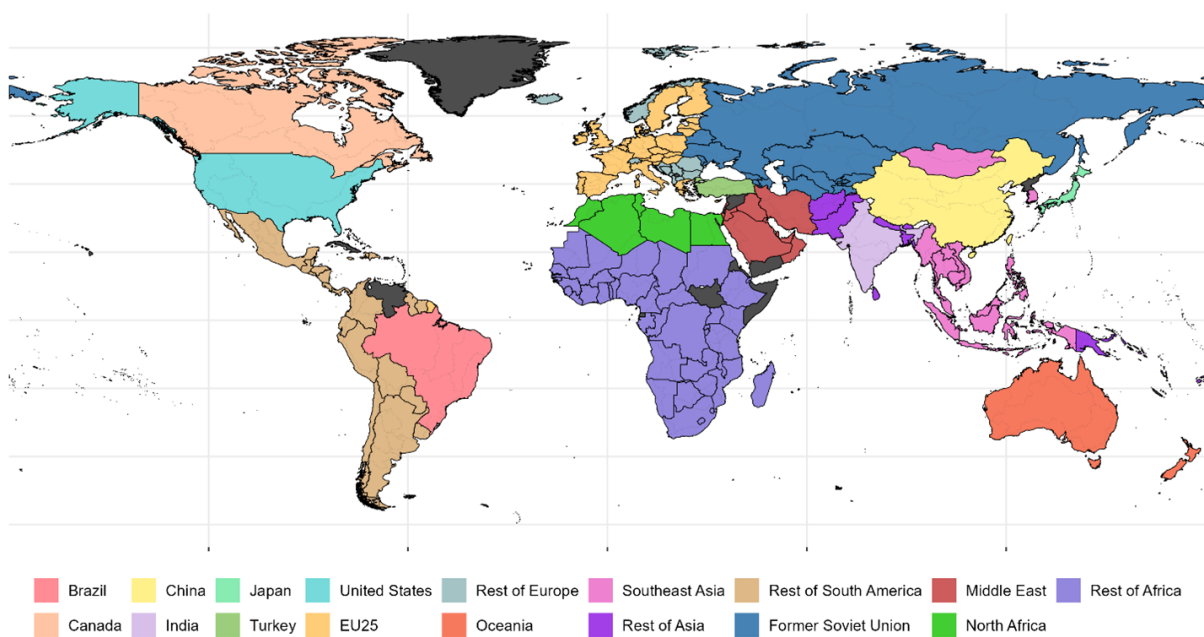


Figure B1: Correspondence between model countries/regions and 17 aggregated regions

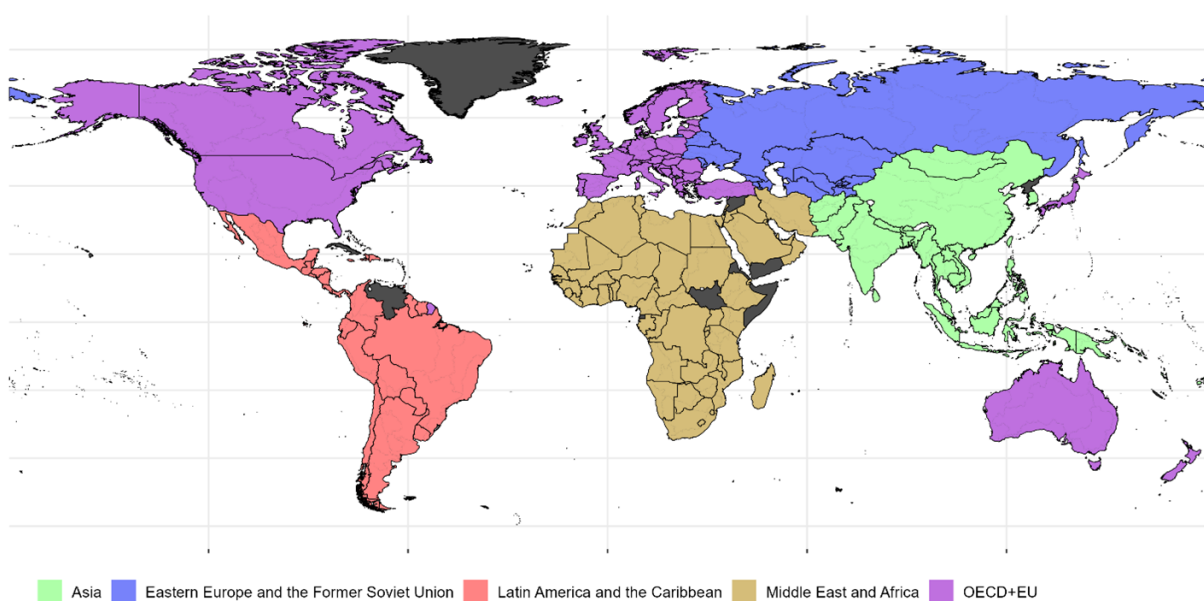


Figure B2: Correspondence between model countries/regions and R5 regions

Appendix C: Income Elasticities

Table C1 summarizes the projected income elasticities for 16 representative countries, selected to cover a wide range of regions and income levels.

Table C1: Projected income elasticities of 16 representative countries by scenario and year

Country	Scenario	Year	wht	rce	crl	mze	str	sgr	swt	stm	alc	pls	nut	ocr	spc	vol	vgt	frt	cmt	rmt	omt	pmt	omt	mlk	dai	egg
AUS	SSP1	2015	0.141	-0.081			-0.181	0.137			0.075		0.036	0.104	0.463	0.342					0.033	0.585	0.033			0.106
AUS	SSP1	2050	0.135	-0.092			-0.187	0.19			0.074		0.036	0.101	0.406	0.31					-0.005	0.55	-0.005			0.102
AUS	SSP1	2100	0.129	-0.103			-0.192	0.165			0.072		0.035	0.098	0.363	0.285					-0.043	0.515	-0.043			0.099
AUS	SSP2	2015	0.141	-0.081			-0.181	0.137			0.075		0.036	0.104	0.463	0.342					0.033	0.585	0.033			0.106
AUS	SSP2	2050	0.136	-0.09			-0.186	0.132			0.074		0.036	0.101	0.413	0.314					0.001	0.555	0.001			0.103
AUS	SSP2	2100	0.13	-0.1			-0.191	0.114			0.072		0.036	0.099	0.373	0.291					-0.033	0.524	-0.033			0.1
AUS	SSP3	2015	0.141	-0.081			-0.181	0.137			0.075		0.036	0.104	0.463	0.342					0.033	0.585	0.033			0.106
AUS	SSP3	2050	0.137	-0.088			-0.185	0.115			0.074		0.036	0.102	0.426	0.321					0.01	0.563	0.01			0.103
AUS	SSP3	2100	0.136	-0.089			-0.186	0.188			0.074		0.036	0.102	0.418	0.317					0.005	0.559	0.005			0.103
AUS	SSP4	2015	0.141	-0.081			-0.181	0.137			0.075		0.036	0.104	0.463	0.342					0.033	0.585	0.033			0.106
AUS	SSP4	2050	0.135	-0.092			-0.187	0.198			0.074		0.036	0.101	0.405	0.31					-0.005	0.55	-0.005			0.102
AUS	SSP4	2100	0.13	-0.101			-0.191	0.114			0.072		0.036	0.098	0.37	0.289					-0.037	0.521	-0.037			0.1
AUS	SSP5	2015	0.141	-0.081			-0.181	0.137			0.075		0.036	0.104	0.463	0.342					0.033	0.585	0.033			0.106
AUS	SSP5	2050	0.134	-0.093			-0.188	0.131			0.073		0.036	0.1	0.4	0.307					-0.009	0.546	-0.009			0.102
AUS	SSP5	2100	0.125	-0.111			-0.196	0.114			0.071		0.035	0.096	0.342	0.271					-0.067	0.493	-0.067			0.097
BRA	SSP1	2015	0.189	0.079			-0.112	0.088			0.088		0.015	0.141	0.755	0.365					0.098	0.456	0.098			0.142
BRA	SSP1	2050	0.177	0.07			-0.113	0.085			0.085		0.015	0.133	0.583	0.319					0.041	0.443	0.041			0.134
BRA	SSP1	2100	0.16	0.06			-0.113	0.08			0.08		0.015	0.124	0.451	0.273					-0.055	0.42	-0.055			0.124
BRA	SSP2	2015	0.189	0.079			-0.112	0.088			0.088		0.015	0.141	0.755	0.365					0.098	0.456	0.098			0.142
BRA	SSP2	2050	0.178	0.071			-0.113	0.085			0.085		0.015	0.134	0.598	0.323					0.048	0.445	0.048			0.135
BRA	SSP2	2100	0.161	0.06			-0.113	0.081			0.081		0.015	0.124	0.459	0.276					-0.047	0.422	-0.047			0.125
BRA	SSP3	2015	0.189	0.079			-0.112	0.088			0.088		0.015	0.141	0.755	0.365					0.098	0.456	0.098			0.142
BRA	SSP3	2050	0.18	0.073			-0.113	0.085			0.085		0.015	0.135	0.623	0.331					0.058	0.447	0.058			0.136
BRA	SSP3	2100	0.171	0.067			-0.114	0.083			0.083		0.015	0.13	0.532	0.303					0.013	0.437	0.013			0.131
BRA	SSP4	2015	0.189	0.079			-0.112	0.088			0.088		0.015	0.141	0.755	0.365					0.098	0.456	0.098			0.142
BRA	SSP4	2050	0.178	0.071			-0.113	0.085			0.085		0.015	0.134	0.596	0.323					0.047	0.444	0.047			0.135
BRA	SSP4	2100	0.161	0.06			-0.113	0.081			0.081		0.015	0.124	0.455	0.275					-0.051	0.421	-0.051			0.125
BRA	SSP5	2015	0.189	0.079			-0.112	0.088			0.088		0.015	0.141	0.755	0.365					0.098	0.456	0.098			0.142
BRA	SSP5	2050	0.175	0.069			-0.113	0.084			0.084		0.015	0.132	0.563	0.313					0.031	0.441	0.031			0.133
BRA	SSP5	2100	0.155	0.056			-0.113	0.079			0.079		0.015	0.12	0.419	0.261					-0.097	0.409	-0.097			0.121
CHN	SSP1	2015	0.164	0.05			-0.348	0.252			0.252		0.033	0.344	0.199	0.144					0.108	0.913	0.108			0.288
CHN	SSP1	2050	0.152	0.044			-0.406	0.225			0.225		0.033	0.296	0.181	0.136					0.063	0.792	0.063			0.253
CHN	SSP1	2100	0.143	0.04			-0.476	0.209			0.209		0.032	0.269	0.17	0.13					0.029	0.719	0.029			0.233
CHN	SSP2	2015	0.164	0.05			-0.348	0.252			0.252		0.033	0.344	0.199	0.144					0.108	0.913	0.108			0.288
CHN	SSP2	2050	0.153	0.044			-0.4	0.227			0.227		0.033	0.3	0.183	0.136					0.067	0.801	0.067			0.256
CHN	SSP2	2100	0.144	0.04			-0.466	0.21			0.21		0.032	0.272	0.171	0.131					0.033	0.727	0.033			0.235
CHN	SSP3	2015	0.164	0.05			-0.348	0.252			0.252		0.033	0.344	0.199	0.144					0.108	0.913	0.108			0.288
CHN	SSP3	2050	0.155	0.045			-0.389	0.231			0.231		0.033	0.306	0.185	0.138					0.074	0.818	0.074			0.261
CHN	SSP3	2100	0.151	0.044			-0.408	0.224			0.224		0.033	0.295	0.181	0.135					0.062	0.788	0.062			0.252
CHN	SSP4	2015	0.164	0.05			-0.348	0.252			0.252		0.033	0.344	0.199	0.144					0.108	0.913	0.108			0.288
CHN	SSP4	2050	0.152	0.044			-0.403	0.226			0.226		0.033	0.298	0.182	0.136					0.065	0.796	0.065			0.255
CHN	SSP4	2100	0.144	0.04			-0.467	0.21			0.21		0.032	0.272	0.171	0.131					0.032	0.726	0.032			0.235
CHN	SSP5	2015	0.164	0.05			-0.348	0.252			0.252		0.033	0.344	0.199	0.144					0.108	0.913	0.108			0.288
CHN	SSP5	2050	0.15	0.043			-0.416	0.222			0.222		0.033	0.291	0.179	0.135					0.057	0.778	0.057			0.249
CHN	SSP5	2100	0.138	0.038			-0.539	0.2			0.2		0.032	0.256	0.165	0.128					0.009	0.682	0.009			0.223
EGY	SSP1	2015	0.147	-0.008			-0.185	0.14			0.14		0.027	0.438	0.211	0.164					0.523	0.866	0.523			0.381
EGY	SSP1	2050	0.133	-0.021			-0.198	0.129			0.129		0.027	0.348	0.187	0.15					0.318	0.717	0.318			0.31
EGY	SSP1	2100	0.12	-0.032			-0.209	0.12			0.12		0.026	0.294	0.169	0.139					0.177	0.631	0.177			0.266
EGY	SSP2	2015	0.147	-0.008			-0.185	0.14			0.14		0.027	0.438	0.211	0.164					0.523	0.866	0.523			0.381
EGY	SSP2	2050	0.135	-0.019			-0.196	0.13			0.13		0.027	0.36	0.191	0.152					0.346	0.736	0.346			0.32
EGY	SSP2	2100	0.123	-0.03			-0.207	0.121			0.121		0.027	0.303	0.172	0.141					0.203	0.646	0.203			0.274
EGY	SSP3	2015	0.147	-0.008			-0.185	0.14			0.14		0.027	0.438	0.211	0.164					0.523	0.866	0.523			0.381
EGY	SSP3	2050	0.139	-0.016			-0.193	0.133			0.133		0.027	0.378	0.196	0.156					0.388	0.765	0.388			0.335
EGY	SSP3	2100	0.134	-0.02			-0.197	0.129			0.129		0.027	0.352	0.188	0.151					0.329	0.724	0.329			0.314
EGY	SSP4	2015	0.147	-0.008			-0.185	0.14			0.14		0.027	0.438	0.211	0.164					0.523	0.866	0.523			0.381
EGY	SSP4	2050	0.137	-0.017			-0.194	0.132			0.132		0.027	0.371	0.194	0.154					0.373	0.754	0.373			0.329
EGY	SSP4	2100	0.127	-0.026			-0.202	0.125			0.125		0.027	0.323	0.179	0.145					0.257	0.678	0.257			0.29
EGY	SSP5	2015	0.147	-0.008			-0.185	0.14			0.14		0.027	0.438	0.211	0.164					0.523	0.866	0.523			0.381
EGY	SSP5	2050	0.131	-0.023			-0.199	0.127			0.127		0.027	0.339	0.184	0.149					0.297	0.704	0.297			0.303
EGY	SSP5	2100	0.116	-0.036			-0.213	0.117			0.117		0.026	0.279	0.163	0.136					0.129	0.604	0.129			

Table C1: Projected income elasticities by country, scenario, and year (continued)

Country	Scenario	Year	wht	rce	crl	mze	str	sgr	swt	stm	alc	pls	nut	ocr	spc	vol	vgt	frt	cmt	rmt	omt	pmt	omt	mlk	dai	egg
DEU	SSP2	2015	0.139	-0.383		-0.119	0.118			0.06			0.033	0.116	0.406	0.322				0.163	1.036	0.163			0.099	
DEU	SSP2	2050	0.135	-0.427		-0.119	0.091			0.059			0.033	0.113	0.379	0.304				0.138	0.958	0.138			0.097	
DEU	SSP2	2100	0.131	-0.491		-0.119	0.103			0.059			0.033	0.111	0.353	0.287				0.11	0.884	0.11			0.095	
DEU	SSP3	2015	0.139	-0.383		-0.119	0.118			0.06			0.033	0.116	0.406	0.322				0.163	1.036	0.163			0.099	
DEU	SSP3	2050	0.136	-0.411		-0.119	0.1			0.06			0.033	0.114	0.387	0.31				0.146	0.982	0.146			0.098	
DEU	SSP3	2100	0.137	-0.405		-0.119	0.099			0.06			0.033	0.114	0.391	0.312				0.149	0.993	0.149			0.098	
DEU	SSP4	2015	0.139	-0.383		-0.119	0.118			0.06			0.033	0.116	0.406	0.322				0.163	1.036	0.163			0.099	
DEU	SSP4	2050	0.135	-0.435		-0.119	0.091			0.059			0.033	0.113	0.375	0.302				0.133	0.946	0.133			0.097	
DEU	SSP4	2100	0.131	-0.495		-0.119	0.079			0.059			0.033	0.11	0.351	0.287				0.109	0.881	0.109			0.095	
DEU	SSP5	2015	0.139	-0.383		-0.119	0.118			0.06			0.033	0.116	0.406	0.322				0.163	1.036	0.163			0.099	
DEU	SSP5	2050	0.134	-0.447		-0.119	0.09			0.059			0.033	0.112	0.369	0.298				0.128	0.931	0.128			0.096	
DEU	SSP5	2100	0.126	-0.616		-0.118	0.091			0.058			0.033	0.108	0.325	0.269				0.079	0.807	0.079			0.093	
IND	SSP1	2015	0.18	0.058		-0.325	0.177			0.177			0.014	0.254	0.659	0.25				3.346	3.655	3.346			0.25	
IND	SSP1	2050	0.163	0.049		-0.412	0.159			0.159			0.014	0.218	0.472	0.219				1.214	1.583	1.214			0.215	
IND	SSP1	2100	0.15	0.043		-0.564	0.147			0.147			0.014	0.197	0.391	0.201				0.785	1.203	0.785			0.194	
IND	SSP2	2015	0.18	0.058		-0.325	0.177			0.177			0.014	0.254	0.659	0.25				3.346	3.655	3.346			0.25	
IND	SSP2	2050	0.165	0.05		-0.4	0.16			0.16			0.014	0.221	0.485	0.221				1.299	1.661	1.299			0.218	
IND	SSP2	2100	0.152	0.044		-0.53	0.148			0.148			0.014	0.2	0.401	0.203				0.833	1.244	0.833			0.197	
IND	SSP3	2015	0.18	0.058		-0.325	0.177			0.177			0.014	0.254	0.659	0.25				3.346	3.655	3.346			0.25	
IND	SSP3	2050	0.167	0.051		-0.383	0.163			0.163			0.014	0.227	0.508	0.226				1.455	1.807	1.455			0.223	
IND	SSP3	2100	0.162	0.049		-0.418	0.158			0.158			0.014	0.216	0.465	0.217				1.174	1.546	1.174			0.213	
IND	SSP4	2015	0.18	0.058		-0.325	0.177			0.177			0.014	0.254	0.659	0.25				3.346	3.655	3.346			0.25	
IND	SSP4	2050	0.166	0.051		-0.389	0.162			0.162			0.014	0.225	0.499	0.224				1.39	1.746	1.39			0.221	
IND	SSP4	2100	0.156	0.046		-0.471	0.152			0.152			0.014	0.207	0.426	0.209				0.96	1.354	0.96			0.204	
IND	SSP5	2015	0.18	0.058		-0.325	0.177			0.177			0.014	0.254	0.659	0.25				3.346	3.655	3.346			0.25	
IND	SSP5	2050	0.161	0.048		-0.425	0.157			0.157			0.014	0.215	0.458	0.216				1.136	1.511	1.136			0.212	
IND	SSP5	2100	0.145	0.04		-0.713	0.142			0.142			0.014	0.189	0.367	0.195				0.67	1.108	0.67			0.187	
IDN	SSP1	2015	0.316	0.069		-0.264	0.136			0.219			0.032	0.36	0.89	0.257				0.75	1.167	0.75			0.921	
IDN	SSP1	2050	0.28	0.063		-0.295	0.083			0.197			0.031	0.306	0.635	0.233				0.498	0.946	0.498			0.652	
IDN	SSP1	2100	0.256	0.057		-0.326	0.181			0.184			0.031	0.275	0.528	0.218				0.374	0.845	0.374			0.54	
IDN	SSP2	2015	0.316	0.069		-0.264	0.136			0.219			0.032	0.36	0.89	0.257				0.75	1.167	0.75			0.921	
IDN	SSP2	2050	0.284	0.063		-0.291	0.083			0.2			0.031	0.312	0.657	0.235				0.521	0.966	0.521			0.674	
IDN	SSP2	2100	0.26	0.058		-0.321	0.202			0.186			0.031	0.28	0.543	0.22				0.393	0.859	0.393			0.556	
IDN	SSP3	2015	0.316	0.069		-0.264	0.136			0.219			0.032	0.36	0.89	0.257				0.75	1.167	0.75			0.921	
IDN	SSP3	2050	0.291	0.065		-0.284	0.084			0.204			0.031	0.321	0.695	0.24				0.561	1	0.561			0.715	
IDN	SSP3	2100	0.283	0.063		-0.292	0.113			0.199			0.031	0.31	0.649	0.234				0.512	0.958	0.512			0.666	
IDN	SSP4	2015	0.316	0.069		-0.264	0.136			0.219			0.032	0.36	0.89	0.257				0.75	1.167	0.75			0.921	
IDN	SSP4	2050	0.288	0.064		-0.287	0.082			0.202			0.031	0.317	0.679	0.238				0.544	0.986	0.544			0.698	
IDN	SSP4	2100	0.269	0.06		-0.308	0.218			0.191			0.031	0.291	0.581	0.226				0.437	0.896	0.437			0.595	
IDN	SSP5	2015	0.316	0.069		-0.264	0.136			0.219			0.032	0.36	0.89	0.257				0.75	1.167	0.75			0.921	
IDN	SSP5	2050	0.276	0.062		-0.3	0.082			0.195			0.031	0.3	0.613	0.23				0.473	0.926	0.473			0.628	
IDN	SSP5	2100	0.245	0.055		-0.348	0.19			0.177			0.031	0.262	0.487	0.211				0.32	0.801	0.32			0.497	
JPN	SSP1	2015	0.159	0.054		-0.393	0.069			0.111			0.021	0.213	0.65	0.418				0.231	1.037	0.231			0.211	
JPN	SSP1	2050	0.153	0.051		-0.426	0.07			0.108			0.021	0.202	0.56	0.38				0.184	0.938	0.184			0.201	
JPN	SSP1	2100	0.146	0.048		-0.474	0.085			0.105			0.021	0.191	0.493	0.348				0.141	0.851	0.141			0.19	
JPN	SSP2	2015	0.159	0.054		-0.393	0.069			0.111			0.021	0.213	0.65	0.418				0.231	1.037	0.231			0.211	
JPN	SSP2	2050	0.154	0.052		-0.42	0.194			0.109			0.021	0.204	0.574	0.386				0.192	0.954	0.192			0.202	
JPN	SSP2	2100	0.148	0.049		-0.46	0.083			0.106			0.021	0.194	0.508	0.356				0.152	0.872	0.152			0.193	
JPN	SSP3	2015	0.159	0.054		-0.393	0.069			0.111			0.021	0.213	0.65	0.418				0.231	1.037	0.231			0.211	
JPN	SSP3	2050	0.156	0.053		-0.409	0.2			0.11			0.021	0.207	0.6	0.397				0.206	0.984	0.206			0.206	
JPN	SSP3	2100	0.156	0.053		-0.407	0.154			0.11			0.021	0.207	0.604	0.399				0.208	0.988	0.208			0.206	
JPN	SSP4	2015	0.159	0.054		-0.393	0.069			0.111			0.021	0.213	0.65	0.418				0.231	1.037	0.231			0.211	
JPN	SSP4	2050	0.153	0.051		-0.427	0.07			0.108			0.021	0.202	0.56	0.38				0.184	0.937	0.184			0.2	
JPN	SSP4	2100	0.148	0.049		-0.464	0.083			0.106			0.021	0.193	0.504	0.354				0.149	0.866	0.149			0.192	
JPN	SSP5	2015	0.159	0.054		-0.393	0.069			0.111			0.021	0.213	0.65	0.418				0.231	1.037	0.231			0.211	
JPN	SSP5	2050	0.151	0.051		-0.436	0.077			0.108			0.021	0.199	0.543	0.372				0.174	0.916	0.174			0.198	
JPN	SSP5	2100	0.142	0.046		-0.524	0.091			0.103			0.021	0.184	0.452	0.328				0.111	0.794	0.111			0.183	
MEX	SSP1	2015	0.341	0.174		-0.042	0.086			0.086			0.024	0.247	0.577	0.44				0.219	0.64	0.219			0.172	
MEX	SSP1	2050	0.3	0.133		-0.04	0.084			0.084			0.024	0.226	0.476	0.378				0.148	0.599	0.148			0.161	
MEX	SSP1	2100	0.269	0.101		-0.038	0.081			0.081			0.024	0.209	0.411	0.335				0.084	0.563	0.084			0.152	
MEX	SSP2	2015	0.341	0.174		-0.042	0.086			0.086			0.024	0.247	0.577	0.44				0.219	0.64	0.219			0.172	
MEX	SSP2	2050	0.306	0.139		-0.04	0.084																			

Table C1: Projected income elasticities by country, scenario, and year (continued)

Country	Scenario	Year	wht	rce	crl	mze	str	sgr	swt	stm	alc	pls	nut	ocr	spc	vol	vgt	frt	cmt	rmt	omt	pmt	omt	mlk	dai	egg
MEX	SSP3	2015	0.341	0.174		-0.042	0.086		0.086				0.024	0.247	0.577	0.44			0.219	0.64	0.219		0.172			
MEX	SSP3	2050	0.317	0.149		-0.041	0.085		0.085				0.024	0.234	0.514	0.402			0.178	0.616	0.178		0.166			
MEX	SSP3	2100	0.313	0.146		-0.041	0.084		0.084				0.024	0.232	0.505	0.396			0.171	0.612	0.171		0.165			
MEX	SSP4	2015	0.341	0.174		-0.042	0.086		0.086				0.024	0.247	0.577	0.44			0.219	0.64	0.219		0.172			
MEX	SSP4	2050	0.306	0.139		-0.04	0.084		0.084				0.024	0.229	0.489	0.386			0.159	0.605	0.159		0.163			
MEX	SSP4	2100	0.273	0.106		-0.038	0.081		0.081				0.024	0.211	0.42	0.341			0.094	0.569	0.094		0.154			
MEX	SSP5	2015	0.341	0.174		-0.042	0.086		0.086				0.024	0.247	0.577	0.44			0.219	0.64	0.219		0.172			
MEX	SSP5	2050	0.293	0.126		-0.04	0.083		0.083				0.024	0.222	0.46	0.368			0.135	0.591	0.135		0.159			
MEX	SSP5	2100	0.253	0.085		-0.037	0.08		0.08				0.024	0.2	0.381	0.315			0.045	0.542	0.045		0.148			
NPL	SSP1	2015	0.228	0.079		-0.195	0.479		0.479				0.016	0.286	0.462	0.346			1.868	2.259	1.868		0.325			
NPL	SSP1	2050	0.206	0.069		-0.224	0.369		0.369				0.015	0.241	0.358	0.284			0.903	1.336	0.903		0.269			
NPL	SSP1	2100	0.182	0.057		-0.265	0.298		0.298				0.015	0.207	0.291	0.239			0.564	1.034	0.564		0.227			
NPL	SSP2	2015	0.228	0.079		-0.195	0.479		0.479				0.016	0.286	0.462	0.346			1.868	2.259	1.868		0.325			
NPL	SSP2	2050	0.209	0.07		-0.219	0.381		0.381				0.015	0.247	0.37	0.291			0.975	1.403	0.975		0.275			
NPL	SSP2	2100	0.185	0.059		-0.259	0.305		0.305				0.015	0.21	0.298	0.244			0.594	1.06	0.594		0.231			
NPL	SSP3	2015	0.228	0.079		-0.195	0.479		0.479				0.016	0.286	0.462	0.346			1.868	2.259	1.868		0.325			
NPL	SSP3	2050	0.215	0.073		-0.212	0.405		0.405				0.015	0.257	0.393	0.305			1.128	1.545	1.128		0.288			
NPL	SSP3	2100	0.203	0.067		-0.228	0.358		0.358				0.015	0.236	0.348	0.277			0.846	1.285	0.846		0.263			
NPL	SSP4	2015	0.228	0.079		-0.195	0.479		0.479				0.016	0.286	0.462	0.346			1.868	2.259	1.868		0.325			
NPL	SSP4	2050	0.212	0.072		-0.215	0.395		0.395				0.015	0.253	0.383	0.299			1.058	1.48	1.058		0.283			
NPL	SSP4	2100	0.192	0.062		-0.246	0.324		0.324				0.015	0.22	0.315	0.256			0.677	1.133	0.677		0.243			
NPL	SSP5	2015	0.228	0.079		-0.195	0.479		0.479				0.016	0.286	0.462	0.346			1.868	2.259	1.868		0.325			
NPL	SSP5	2050	0.202	0.067		-0.229	0.355		0.355				0.015	0.235	0.346	0.276			0.831	1.271	0.831		0.261			
NPL	SSP5	2100	0.176	0.054		-0.282	0.282		0.282				0.015	0.198	0.276	0.228			0.494	0.973	0.494		0.217			
NGA	SSP1	2015	0.429	0.296		-0.064	0.247		0.247				0.017	0.286	0.873	0.362			1.985	2.945	1.985		1.875			
NGA	SSP1	2050	0.38	0.252		-0.065	0.225		0.225				0.017	0.258	0.66	0.319			1.126	1.972	1.126		1.127			
NGA	SSP1	2100	0.31	0.193		-0.064	0.194		0.194				0.017	0.218	0.469	0.263			0.635	1.369	0.635		0.699			
NGA	SSP2	2015	0.429	0.296		-0.064	0.247		0.247				0.017	0.286	0.873	0.362			1.985	2.945	1.985		1.875			
NGA	SSP2	2050	0.391	0.262		-0.065	0.23		0.23				0.017	0.264	0.7	0.328			1.252	2.121	1.252		1.239			
NGA	SSP2	2100	0.325	0.205		-0.064	0.201		0.201				0.017	0.227	0.502	0.274			0.713	1.467	0.713		0.764			
NGA	SSP3	2015	0.429	0.296		-0.064	0.247		0.247				0.017	0.286	0.873	0.362			1.985	2.945	1.985		1.875			
NGA	SSP3	2050	0.404	0.274		-0.064	0.236		0.236				0.017	0.272	0.754	0.34			1.441	2.339	1.441		1.406			
NGA	SSP3	2100	0.375	0.248		-0.065	0.223		0.223				0.017	0.255	0.645	0.315			1.082	1.919	1.082		1.087			
NGA	SSP4	2015	0.429	0.296		-0.064	0.247		0.247				0.017	0.286	0.873	0.362			1.985	2.945	1.985		1.875			
NGA	SSP4	2050	0.401	0.271		-0.065	0.235		0.235				0.017	0.27	0.739	0.337			1.384	2.274	1.384		1.356			
NGA	SSP4	2100	0.349	0.225		-0.065	0.211		0.211				0.017	0.24	0.563	0.293			0.862	1.652	0.862		0.893			
NGA	SSP5	2015	0.429	0.296		-0.064	0.247		0.247				0.017	0.286	0.873	0.362			1.985	2.945	1.985		1.875			
NGA	SSP5	2050	0.372	0.245		-0.065	0.222		0.222				0.017	0.253	0.633	0.312			1.049	1.879	1.049		1.058			
NGA	SSP5	2100	0.293	0.179		-0.063	0.186		0.186				0.017	0.209	0.434	0.25			0.552	1.265	0.552		0.631			
RUS	SSP1	2015	0.127	-0.506		-0.243	0.222		0.08				0.092	0.167	0.502	0.367			0.192	0.698	0.192		0.129			
RUS	SSP1	2050	0.122	-0.627		-0.254	0.177		0.079				0.09	0.16	0.441	0.334			0.146	0.662	0.146		0.124			
RUS	SSP1	2100	0.114	-1.132		-0.273	0.085		0.076				0.086	0.15	0.377	0.296			0.081	0.61	0.081		0.118			
RUS	SSP2	2015	0.127	-0.506		-0.243	0.222		0.08				0.092	0.167	0.502	0.367			0.192	0.698	0.192		0.129			
RUS	SSP2	2050	0.122	-0.613		-0.253	0.194		0.079				0.09	0.16	0.446	0.336			0.15	0.665	0.15		0.124			
RUS	SSP2	2100	0.115	-1.002		-0.271	0.083		0.076				0.087	0.151	0.384	0.3			0.089	0.616	0.089		0.119			
RUS	SSP3	2015	0.127	-0.506		-0.243	0.222		0.08				0.092	0.167	0.502	0.367			0.192	0.698	0.192		0.129			
RUS	SSP3	2050	0.123	-0.592		-0.252	0.128		0.079				0.09	0.161	0.454	0.341			0.157	0.67	0.157		0.125			
RUS	SSP3	2100	0.12	-0.684		-0.258	0.087		0.078				0.089	0.157	0.425	0.324			0.132	0.65	0.132		0.123			
RUS	SSP4	2015	0.127	-0.506		-0.243	0.222		0.08				0.092	0.167	0.502	0.367			0.192	0.698	0.192		0.129			
RUS	SSP4	2050	0.122	-0.627		-0.254	0.198		0.079				0.09	0.16	0.441	0.334			0.146	0.662	0.146		0.124			
RUS	SSP4	2100	0.115	-1.038		-0.271	0.083		0.076				0.087	0.15	0.382	0.299			0.087	0.614	0.087		0.118			
RUS	SSP5	2015	0.127	-0.506		-0.243	0.222		0.08				0.092	0.167	0.502	0.367			0.192	0.698	0.192		0.129			
RUS	SSP5	2050	0.121	-0.652		-0.256	0.175		0.078				0.089	0.158	0.434	0.329			0.139	0.656	0.139		0.123			
RUS	SSP5	2100	0.109	-2.439		-0.284	0.091		0.075				0.085	0.145	0.352	0.281			0.05	0.585	0.05		0.115			
ZAF	SSP1	2015	0.227	0.088		-0.068	0.146		0.093				0.096	0.284	1.098	0.701			0.192	0.535	0.192		0.324			
ZAF	SSP1	2050	0.21	0.072		-0.067	0.19		0.09				0.093	0.259	0.811	0.57			0.133	0.512	0.133		0.292			
ZAF	SSP1	2100	0.183	0.048		-0.064	0.152		0.085				0.088	0.223	0.569	0.433			0.019	0.473	0.019		0.248			
ZAF	SSP2	2015	0.227	0.088		-0.068	0.146		0.093				0.096	0.284	1.098	0.701			0.192	0.535	0.192		0.324			
ZAF	SSP2	2050	0.211	0.073		-0.068	0.147		0.091				0.093	0.261	0.828	0.578			0.138	0.514	0.138		0.295			
ZAF	SSP2	2100	0.185	0.05		-0.065	0.136		0.086				0.088	0.225	0.582	0.441			0.028	0.476	0.028					

Table C1: Projected income elasticities by country, scenario, and year (continued)

Country	Scenario	Year	wht	rce	crl	mze	str	sgr	swt	stm	alc	pls	nut	ocr	spc	vol	vgt	frt	cmt	rmt	omt	pmt	omt	mlk	dai	egg
ZAF	SSP4	2015	0.227	0.088		-0.068	0.146		0.093				0.096	0.284	1.098	0.701					0.192	0.535	0.192		0.324	
ZAF	SSP4	2050	0.212	0.074		-0.068	0.154		0.091				0.093	0.261	0.831	0.58					0.139	0.514	0.139		0.295	
ZAF	SSP4	2100	0.186	0.05		-0.065	0.146		0.086				0.088	0.227	0.588	0.445					0.033	0.478	0.033		0.252	
ZAF	SSP5	2015	0.227	0.088		-0.068	0.146		0.093				0.096	0.284	1.098	0.701					0.192	0.535	0.192		0.324	
ZAF	SSP5	2050	0.207	0.07		-0.067	0.214		0.09				0.092	0.255	0.779	0.553					0.123	0.508	0.123		0.287	
ZAF	SSP5	2100	0.175	0.041		-0.063	0.132		0.084				0.086	0.214	0.522	0.403					-0.023	0.46	-0.023		0.237	
TUR	SSP1	2015	0.124	-0.295		-0.304	0.222		0.129				0.011	0.129	0.245	0.214					0.582	0.958	0.582		0.117	
TUR	SSP1	2050	0.116	-0.364		-0.336	0.07		0.123				0.011	0.123	0.223	0.197					0.378	0.8	0.378		0.111	
TUR	SSP1	2100	0.112	-0.424		-0.359	0.082		0.119				0.011	0.119	0.212	0.189					0.287	0.738	0.287		0.108	
TUR	SSP2	2015	0.124	-0.295		-0.304	0.222		0.129				0.011	0.129	0.245	0.214					0.582	0.958	0.582		0.117	
TUR	SSP2	2050	0.117	-0.358		-0.334	0.068		0.123				0.011	0.123	0.224	0.198					0.389	0.808	0.389		0.112	
TUR	SSP2	2100	0.112	-0.415		-0.356	0.083		0.12				0.011	0.12	0.213	0.19					0.298	0.746	0.298		0.109	
TUR	SSP3	2015	0.124	-0.295		-0.304	0.222		0.129				0.011	0.129	0.245	0.214					0.582	0.958	0.582		0.117	
TUR	SSP3	2050	0.117	-0.35		-0.33	0.068		0.124				0.011	0.124	0.226	0.2					0.407	0.821	0.407		0.112	
TUR	SSP3	2100	0.113	-0.399		-0.35	0.117		0.12				0.011	0.12	0.216	0.192					0.32	0.76	0.32		0.109	
TUR	SSP4	2015	0.124	-0.295		-0.304	0.222		0.129				0.011	0.129	0.245	0.214					0.582	0.958	0.582		0.117	
TUR	SSP4	2050	0.116	-0.362		-0.335	0.198		0.123				0.011	0.123	0.223	0.197					0.382	0.803	0.382		0.111	
TUR	SSP4	2100	0.112	-0.415		-0.356	0.083		0.12				0.011	0.12	0.213	0.19					0.299	0.746	0.299		0.109	
TUR	SSP5	2015	0.124	-0.295		-0.304	0.222		0.129				0.011	0.129	0.245	0.214					0.582	0.958	0.582		0.117	
TUR	SSP5	2050	0.115	-0.375		-0.341	0.095		0.122				0.011	0.122	0.22	0.195					0.357	0.785	0.357		0.111	
TUR	SSP5	2100	0.108	-0.497		-0.383	0.088		0.116				0.011	0.116	0.203	0.182					0.218	0.696	0.218		0.106	
USA	SSP1	2015	0.138	-0.106		-0.131	0.098		0.057				0.032	0.102	0.432	0.325					0.025	0.535	0.025		0.092	
USA	SSP1	2050	0.134	-0.115		-0.132	0.19		0.056				0.031	0.1	0.396	0.305					-0.011	0.514	-0.011		0.09	
USA	SSP1	2100	0.129	-0.124		-0.132	0.106		0.055				0.031	0.098	0.367	0.288					-0.046	0.494	-0.046		0.088	
USA	SSP2	2015	0.138	-0.106		-0.131	0.098		0.057				0.032	0.102	0.432	0.325					0.025	0.535	0.025		0.092	
USA	SSP2	2050	0.134	-0.113		-0.131	0.194		0.056				0.032	0.1	0.401	0.308					-0.005	0.518	-0.005		0.09	
USA	SSP2	2100	0.131	-0.122		-0.132	0.112		0.056				0.031	0.098	0.375	0.292					-0.036	0.5	-0.036		0.089	
USA	SSP3	2015	0.138	-0.106		-0.131	0.098		0.057				0.032	0.102	0.432	0.325					0.025	0.535	0.025		0.092	
USA	SSP3	2050	0.135	-0.112		-0.131	0.2		0.056				0.032	0.101	0.408	0.312					0.002	0.522	0.002		0.09	
USA	SSP3	2100	0.134	-0.114		-0.131	0.188		0.056				0.032	0.1	0.401	0.307					-0.006	0.517	-0.006		0.09	
USA	SSP4	2015	0.138	-0.106		-0.131	0.098		0.057				0.032	0.102	0.432	0.325					0.025	0.535	0.025		0.092	
USA	SSP4	2050	0.134	-0.115		-0.132	0.198		0.056				0.031	0.1	0.395	0.304					-0.012	0.514	-0.012		0.09	
USA	SSP4	2100	0.13	-0.123		-0.132	0.107		0.056				0.031	0.098	0.371	0.29					-0.04	0.498	-0.04		0.088	
USA	SSP5	2015	0.138	-0.106		-0.131	0.098		0.057				0.032	0.102	0.432	0.325					0.025	0.535	0.025		0.092	
USA	SSP5	2050	0.133	-0.116		-0.132	0.131		0.056				0.031	0.099	0.391	0.302					-0.017	0.511	-0.017		0.09	
USA	SSP5	2100	0.125	-0.134		-0.132	0.109		0.055				0.031	0.096	0.343	0.273					-0.082	0.475	-0.082		0.087	
VNM	SSP1	2015	0.552	0.068		-0.521	0.287		0.287				0.014	0.669	0.455	0.119					0.183	0.974	0.183		0.72	
VNM	SSP1	2050	0.432	0.06		-0.864	0.244		0.244				0.013	0.484	0.358	0.116					0.124	0.832	0.124		0.511	
VNM	SSP1	2100	0.385	0.053		-2.52	0.226		0.226				0.013	0.426	0.323	0.123					0.092	0.772	0.092		0.448	
VNM	SSP2	2015	0.552	0.068		-0.521	0.287		0.287				0.014	0.669	0.455	0.119					0.183	0.974	0.183		0.72	
VNM	SSP2	2050	0.438	0.061		-0.814	0.246		0.246				0.013	0.493	0.363	0.116					0.128	0.84	0.128		0.521	
VNM	SSP2	2100	0.389	0.055		-4.084	0.227		0.227				0.013	0.43	0.325	0.121					0.094	0.776	0.094		0.452	
VNM	SSP3	2015	0.552	0.068		-0.521	0.287		0.287				0.014	0.669	0.455	0.119					0.183	0.974	0.183		0.72	
VNM	SSP3	2050	0.453	0.062		-0.735	0.252		0.252				0.013	0.513	0.375	0.117					0.137	0.858	0.137		0.544	
VNM	SSP3	2100	0.429	0.06		-0.888	0.243		0.243				0.013	0.48	0.355	0.116					0.122	0.828	0.122		0.507	
VNM	SSP4	2015	0.552	0.068		-0.521	0.287		0.287				0.014	0.669	0.455	0.119					0.183	0.974	0.183		0.72	
VNM	SSP4	2050	0.446	0.062		-0.77	0.249		0.249				0.013	0.503	0.369	0.116					0.132	0.849	0.132		0.532	
VNM	SSP4	2100	0.398	0.058		-1.916	0.23		0.23				0.013	0.439	0.331	0.116					0.1	0.786	0.1		0.462	
VNM	SSP5	2015	0.552	0.068		-0.521	0.287		0.287				0.014	0.669	0.455	0.119					0.183	0.974	0.183		0.72	
VNM	SSP5	2050	0.423	0.06		-0.953	0.24		0.24				0.013	0.471	0.35	0.116					0.118	0.82	0.118		0.498	
VNM	SSP5	2100	0.356	0.044		-2.813	0.217		0.217				0.013	0.398	0.306	0.138					0.072	0.739	0.072		0.418	